CHAPTER ONE

1.0 FOOD TESTS

These are tests that can be used to determine whether specific food substances are present or absent in a given food samples either in solution or in solid state.

The major foods eaten by animals are of two forms. These include: - organic compounds and inorganic compounds.

The main organic compounds are: - **carbohydrates, proteins, lipids and vitamins**.
The carbohydrates are divided into the following

(i) Monosaccharides for example glucose, fructose and galactose.
(ii) Disaccharides for example lactose, maltose and sucrose.
(iii) Polysaccharides for example starch, cellulose in plants and glycogen in animals, chitin in exoskeleton of arthropods.

All monosaccharides (simple sugars) and disaccharides (complex sugars) except sucrose are called reducing sugars because they have got an active reducing group. Sucrose is called non-reducing sugar.

The food substance tested during food tests include

(i) Starch
(ii) Reducing sugars
(iii) Non-reducing sugars
(iv) Proteins
(v) Lipid (fats and oils)
(vi) Vitamin C (Ascorbic acid)

When carrying out food test on any food substance it is very important to note the following:-

- The test reagent (i.e. type of food substance it tests, its nature and colour).
- The test procedure (i.e. proportionality, order of addition and conditions like heating, cooling and shaking).
- The colour change expected
- The colour of the original/test solution before adding the reagent i.e. turbid, colourless, milky, cloudy e.t.c.
- The nature of the test and resultant substance i.e. solution or precipitate.

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1.1 THE TABLE BELOW SHOWS THE TEST REAGENTS AND THEIR COLOUR

REAGENT	ORIGINAL COLOUR
Iodine solution	Brown
Benedict's/Fehling's solution	Blue
Millon's reagent	Colourless
Copper (II) sulphate solution	Blue
Dilute Hydrochloric acid	Colourless
Sodium hydroxide solution	Colourless
Ethanol	Colourless
Distilled water	Colourless
DCPIP solution	Blue

1.2 THE TABLE BELOW SHOWS THE FOOD TYPE AND THE TEST REAGENT USED

FOOD TYPE	TEST REAGENT
Starch	Iodine solution
Reducing sugar	Benedict's solution/Fehling's solution
Non-reducing sugar	Benedict's solution/Fehling's solution, Dilute hydrochloric acid and sodium hydroxide solution
Proteins	Millon's reagent
	Copper II sulphate solution and sodium hydroxide solution (Biuret's test)
Lipids	Ethanol and water
Vitamin C	DCPIP (Dichlorophenolindophenol) solution

1.3 REDUCING SUGARS (Benedict's test/ Fehling's Test)

Test	Observation	Deduction
To $1cm^3$ of the test solution, add $1cm^3$ of Benedict's solution and boil.	A colourless solution changes to blue solution, then to green solution, yellow precipitate, orange precipitate and finally brown precipitate on boiling.	Reducing sugar present.
	Colourless solution turns blue solution which persists/ remains on boiling.	Reducing sugar absent.

Note: During the above test, the final colour indicates the quantity of reducing sugars present in a given food sample or solution i.e.

- Blue shows that no reducing sugars are present.
- Green shows that little reducing sugars are present.
- Yellow shows that moderate reducing sugars are present.
- Orange and brown/red shows that much reducing sugars are present.

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1.4 NON REDUCING SUGARS

Test	Observation	Deduction
To $1cm^3$ of the test solution, add $1cm^3$ of dilute hydrochloric acid and boil for 1 minute. Cool under tap water and then add $1cm^3$ of sodium hydroxide solution followed by $1cm^3$ of Benedict's solution and boil again.	A colourless solution changes to blue solution, then to green solution, yellow precipitate, orange precipitate and finally brown precipitate on boiling.	Non-reducing sugar present.
	Colourless solution turns blue solution which persists/ remains on boiling.	Non-reducing sugar absent.

Note:

In testing for non-reducing sugars;

- Dilute hydrochloric acid is added to hydrolyse the non-reducing sugars to reducing sugars.
- The mixture is boiled to increase the rate of hydrolysis reaction and is cooled to stop the hydrolysis reaction.
- Sodium hydroxide or sodium hydrogen carbonate solution is added to neutralize the excess acid added because Benedict's / Fehling's solution does not work in acidic conditions.

1.5 STARCH (Iodine test)

Test	Observation	Deduction
To $1cm^3$ of test solution in a test tube add 2-3 drops of iodine solution.	Milky/ cloudy solution turns to blue-black/black/blue solution.	Starch present.
	Milky/cloudy solution turns to brown solution.	Starch absent.

1.6 LIPIDS

(i) TRANSLUCENT MARK TEST

Test	Observation	Deduction
Rub the substance on a piece of paper (if provided as solid) or add 2 drops of test solution on a piece of paper. Allow to dry, hold the paper against light and observe.	Translucent spot or patch or mark is left on the paper.	Lipids present.
	No translucent mark is observed.	Lipids absent

(ii) SUDAN III TEST

Test	Observation	Deduction
To $1cm^3$ of test solution add $2cm^3$ of distilled water followed by 2-3 drops of Sudan III and shake.	Red droplets are seen in the mixture.	Lipids present.
	Solution remains with the original colour.	Lipids absent

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(iii) EMULSION/ETHANOL TEST

Test	Observation	Deduction
To 1cm^3 of the test solution add 5cm^3 of ethanol and shake vigorously followed by 2cm^3 of distilled water.	Cloudy/ creamy/ milky emulsion/suspension is formed.	Lipids present.
	Mixture remains colourles	Lipids absent

1.7 PROTEINS

(i) MILLON'S TEST

Test	Observation	Deduction
To 1cm^3 of test solution add 4 drops of Millon's reagent and boil for 2 minutes.	A pink or red coagulated mass forms.	Proteins present.
	Colourless solution remains colourless solution	Proteins absent

(ii) BIURET'S TEST

Test	Observation	Deduction
To 1cm^3 of the test solution add 1cm^3 of Sodium hydroxide solution followed 2 drops of copper II sulphate solution and shake slightly.	Turbid/milky solution turns to purple solution.	Proteins present.
	Turbid/milky solution turns into colourless solution turns then to blue precipitate.	Proteins absent.

1.8 VITAMIN C

Test	Observation	Deduction
To 1cm^3 of DCPIP solution in a test tube add the test solution drop wise until in excess.	The blue colour DCPIP solution is decolourised/ discharged i.e. The blue solution turns to colourless solution.	Vitamin C present
	The blue colour of DCPIP solution remains/persists	Vitamin C absent

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1.9 ENZYME ACTION

Enzymes are biological catalysts, proteins in nature that speeds up rate of chemical reactions. Enzymes act on substances known substrates.

1.9.1 Characteristics of enzymes:

- They work on specific substrates.
- They work at specific pH i.e. they are affected by pH changes.

- They work best at an optimum temperature (35°C-40°C).
- They are denatured by excess heat.

1.9.2 The Water Bath

The water bath is an alternative method of heating in food tests or food related experiments like those involving the investigation of the action of enzymes on food substances. It should be noted that in all enzyme controlled reactions, the temperature of water bath should be kept at least between 35-40^{0}C for the whole length of the experiment.

1.9.3 The following are the factors that affect the rate of enzyme action

(i) ***Temperature.***

Enzyme action increases with temperature up the optimum, beyond which enzymes are denatured.

(ii) ***pH.***

Enzymes have specific optimum pH. Any change in pH will affect enzyme action. pH can be acidic, alkaline or neutral.

(iii) ***Concentration of substrate.***

Enzyme activity increases with increase in substrate concentration.

(iv) Concentration of enzymes.

Enzyme activity increases with increase in its concentration.

(v) Co-factor/ co-enzyme

(vi) Inhibitors

These reduce enzyme activity.

1.9.4 HYDROLYSIS OF STARCH USING SALIVARY AMYLASE:

Starch is hydrolyzed by salivary amylase to maltose (reducing sugar) in a suitable pH medium i.e neutral or alkaline medium. In acidic medium starch is not hydrolyzed.

Test	**Observation**	**Deduction**
To 1cm^3 of starch solution, in a test tube, add 1cm^3 of diluted saliva. Warm in a water bath at 30-35^{0}C for 10 minutes. Add a few drops of iodine solution.	The brown colour of iodine solution persists/remains	Starch was hydrolyzed to reducing sugars
To 1cm^3 of starch solution, in a test tube, add 1cm^3 of diluted saliva followed by 10 drops of dilute HCl. Warm in a water bath at 30-35^{0}C for 10 minutes. Add a few drops of iodine solution.	the milky/cloudy turns to blue-black solution	Starch was not hydrolyzed to reducing sugars
To 1cm^3 of starch solution, in a test tube, add 1cm^3 of diluted saliva followed by 10 drops of sodium hydroxide solution. Warm in a water bath at 30-35^{0}C for 10 minutes. Add a few drops of iodine solution.	The brown colour of iodine solution persists/remains	Starch was hydrolyzed to reducing sugars

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1.9.4 HYDROLYSIS OF STARCH BY OTHER ENZYMES (not salivary amylase):

These enzymes can be amylase in maize, diastase in barley. These enzymes are extracted from seeds by soaking different sets of seeds for example maize, for different times that is to say two days, seven days and ten days. They are then ground and water is added into them, to get fine extracts.

A control is set up by using an extract prepared from un soaked seeds.

These are then tested with Benedict's solution for reducing sugars.

In each of the extracts a sample is tested for reducing sugar using benedict's solution. The test will be positive soaked in water for a longer time and were germinating. This is because germination involves mobilizing the food reserves by the enzyme activity to soluble usable forms i.e. breaking down starch to maltose.

In unsoaked seeds, the Benedict's test will be negative i.e. starch is not broken down.

1.9.5 HYDROLYSIS OF SUCROSE BY INVERTASE

Invertase is the enzyme found in yeast and it brings about the conversion of sucrose to glucose and fructose which are reducing sugars.

Test	Observation	Deduction
To $1cm^3$ of sucrose solution add $1cm^3$ of yeast suspension and incubate the mixture in a water bath maintained at temperature of 35^oC to 40^0C for 15-20 minutes and then add $1cm^3$ Benedict's solution and boil.	A colourless solution changes to blue solution, then to green solution, yellow precipitate, orange precipitate and finally brown precipitate on boiling.	Non-reducing sugars (sucrose) was hydrolysed to reducing sugars

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1.9.6 ALCOHOLIC FERMENTATION USING ENZYMES IN YEAST

Procedure

Mix glucose and yeast in a test tube and pour a layer of oil over it, this ensures anaerobic conditions. Fit it with a delivery tube fitted to another test tube containing lime water or indicator solution to test carbon dioxide evolved.

For positive results (in presence of carbon dioxide) if lime water is used

Observation

Solutions changes milky or cloudy

If indicators are used like bromethymol blue- changes colour from blue to yellow

Cresol red -changes colour from red to yellow

1.9.7 HYDROLYSIS OF PROTEINS BY THE ENZYME PEPSIN

Proteins are hydrolyzed to amino acids by pepsin in acidic solution. However, they can also be hydrolyzed by Trypsin but an alkaline medium.

Test	Observation	Deduction
To $1cm^3$ of protein solution add $1cm^3$ of pepsin solution followed by 2-3 drops of dilute HCl and incubate the mixture in a water bath maintained at temperature of 35^oC to 40^0C for 15-20 minutes.	The turbid/milky/cloudy solution turns to colourless solution i.e. the turbidity/ milkness/cloudness disappears.	Proteins were hydrolysed.

1.9.8 HYDROLYSIS OF HYDROGEN PEROXIDE BY THE ENZYME CATALASE

H_2O_2 is a toxic (poisonous) substance produced as a result of metabolic activity in the cells of both plant and animal tissues. These tissues contain an enzyme catalase which is capable of decomposing hydrogen peroxide (H_2O_2) into water and oxygen which are non-toxic.

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

Highly active tissues like liver cells show a very fast reaction compared to low active ones, and generally animal tissues show greater activity than plant tissues.

When a piece of potato tuber or liver is dropped in a solution of hydrogen peroxide, gas bubbles will be seen coming out of the potato or liver surface. This gas can be collected and will relight a glowing splint indicating that it is oxygen.

This reaction does not occur in dead tissues e.g. boiled tissues. If boiled potato tuber or liver tissue instead of unboiled, no gas bubbles is given out. This is because boiling denatures enzymes in the potato or liver tissue.

If the potato tuber or liver tissue is cut into small pieces or crushed and dropped in the hydrogen peroxide solution, the reaction will be much faster as shown with faster evolution of gas bubbles. This is because crushed or small pieces of potato tuber or liver tissues have large surface area for enzyme reaction.

The reaction is fastest when pH is neutral, is moderate in alkaline, and very slow or stops in acidic medium.

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1.10 WORKED EXAMPLES:

Example 1.

(a) You are provided with solution **N**. Carry out the following tests to identify the food substance in the solution. Record your observation and deductions, in the table below.

TEST	OBSERVATION	DEDUCTION
(i) To $1cm^3$ of solution **N** in a test tube add 2-3 drops of iodine solution	Milky solution turned to blue-black solution	Starch present
(ii) To $1cm^3$ of solution **N** in a test tube add $1cm^3$ of Benedict's solution and boil	Milky solution turned to blue solution and remained blue solution on boiling.	Reducing sugar absent
(iii) To $1cm^3$ of solution **N** add 3-4 drops of dilute hydrochloric acid and boil for one minute cool under tap water and add 3-4 drops of sodium hydroxide. Then add Benedict's solution and boil.	Milky solution turned to blue solution and remained blue solution on boiling.	Non reducing sugar absent
(iv) To $1cm^3$ of solution **N** in a test tube add dilute sodium hydroxide followed by 4 drops of copper sulphate.	Milky solution turned into colourless solution, then to purple solution	Proteins present
(v) To $1cm^3$ **DCPIP** in a test tube add solution **N** drop by drop	Blue solution remained blue solution	Vitamin C absent

(b) Identify the food substances in solution **N**

Starch and Proteins

(c) Name two sources from which the above foods can be obtained.

Sources of starch include cassava, maize, wheat e.t.c

Sources of protein include meat, beans, chicken, eggs, e.t.c

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EXAMPLE 2

a. You are provided with an albumin solution (R), pepsin, dilute hydrochloric acid, sodium hydroxide solution and 4 test tubes. Carry out the tests below

Tests	**Observations**	**Deductions**
(i) To $2cm^3$ of **R** in a test tube 1, add $1cm^3$ of pepsin solution.	The turbid solution remains turbid	Proteins were not hydrolysed by pepsin
(ii) To $2cm^3$ of **R** in a test tube 2, add $1cm^3$ of pepsin solution followed by 2 drops of dilute hydrochloric acid.	The turbid solution becomes clear/colourless solution	Proteins were hydrolysed by pepsin.
(iii) To $2cm^3$ of **R** in a test tube 3, add $1cm^3$ of pepsin solution followed by 2 drops of sodium hydroxide solution	The turbid solution remains turbid	Proteins were not hydrolysed by pepsin
(iv) To $2cm^3$ of **R** in a test tube 4, add $1cm^3$ of boiled pepsin solution followed by 2 drops of dilute hydrochloric acid.	The turbid solution remains turbid	Proteins were not hydrolysed by pepsin

Place all the test tubes in a water bath maintained at a temperature between 35oC-40oC for 20 minutes and then record your observatons.

b. Explain your results in (a) above.

i. Test tube 1

There was no hydrolysis of proteins by pepsin because pepsin does not work in a neutral pH medium.

ii. Test tube 2

Proteins were hydrolysed because the hydrochloric acid provided the optimum pH medium for pepsin enzyme. I.e. pepsin works best in acidic medium.

iii. Test tube 3

There was no hydrolysis of proteins by pepsin because pepsin does not work in alkaline pH medium

iv. Test tube 4

There was no hydrolysis of proteins by pepsin because pepsin/enzyme had been denatured by excessive heat during boiling.

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EXAMPLE 3

You are provided with solution **H.** Carry out the following tests to identify the food substances in solution **H** and investigate the effect of Saliva on the food substances in **H.** Record your observations and deductions in the table below**:**

Tests	Observations	Deductions
(i)To 1cm^3 of **H** in a test tube add 2-3 drops of iodine solution.	Milky solution turned to blue-black solution	Starch present
(ii)To 1cm^3 of **H** in a test tube add 1cm^3 of Benedict's solution and boil for 1 minute.	Milky solution turned to blue solution and remained blue solution on boiling.	Reducing sugar absent

(a) Rinse your mouth with water. Spit 3cm^3 of saliva in a test tube; dilute it with an equal volume of water. Label the solution **X.**

Label 3 test tubes **A, B** and **C**. Add 1cm^3 of **H** to each test tube and treat them as follows:

(i) To test tube **A** add 2cm^3 of solution **X** followed by 2cm^3 of dilute Hydrochloric acid.

(ii) To test tube **B** add 2cm^3 of solution **X** followed by 2cm^3 of dilute Sodium Hydroxide solution.

(iii) To test tube **C** add 2cm^3 of solution **X** and boil the mixture for 1 minute.

Incubate all the 3 test tubes in a warm water bath maintained at temperature between 35º-40ºC for 20 minutes. (*You may continue with other work in the meantime*.) After 20 minutes carry out the following Benedict's tests and record your observations and deductions in the table below.

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Test tubes	Observations	Deductions
A	Milky solution turned to blue solution and remained blue solution on boiling.	Reducing sugar absent
B	Milky solution turned to blue solution, then green precipitate and yellow precipitate on boiling.	Reducing sugar present
C	Milky solution turned to blue solution and remained blue solution on boiling.	Reducing sugar absent

(b) Explain your observations and deductions in the table above

(i) Test tube **A**

Solution H was not hydrolysed to reducing sugars by the active substance in solution X because the active substance in X does not work in acidic pH medium

(ii) Test tube **B**

Solution H was hydrolysed to reducing sugars by the active substance in solution X because the active substance in X does works best in alkaline pH medium

(iii) Test tube **C**

Solution H was not hydrolysed to reducing sugars by the active substance in solution X because the active substance in X was denatured by excessive heat during boiling.

(c) From your observations state the nature of the active substance in Solution **X**

It is an enzyme

(d) State 3 properties of the active substance

It is denatured by excessive heat.

It is affected by pH changes.

It is specific in nature

EXAMPLE 4

You are provided with the following: Hydrogen peroxide solution, Solution X, Solution Z and pieces of liver tissue, litmus papers and test tubes.

a) Carry out the following tests and record your observations and deductions in the table below. Label four test tubes 1, 2, 3 and 4 and place a piece of liver tissues provided in each test tube.

Test tube	**Test**		**Observation**	**Deduction**
1	i)	Add solution **X** to completely cover the piece of liver tissue and add 1cm3 of hydrogen peroxide.	Few bubbles of gas evolved/ Little efferverscence occurs	Slow decomposition of hydrogen peroxide
	ii)	Test the mixture with red and blue litmus paper.	Red litmus turned blue while blue litmus remained blue	The mixture is alkaline
2	i)	Add solution **Z** to completely cover the liver tissue and add 1cm3 of hydrogen peroxide	Few bubbles of gas evolved/ Little efferverscence occurs	Slow decomposition of hydrogen peroxide
	ii)	Test the mixture with red and blue litmus paper.	Blue litmus turned red while red litmus remained red	The mixture is acidic
3	i)	Add distilled water to cover the liver tissue and 1cm3 of hydrogen peroxide.	Bubbles of gas are liberated vigorously/ fast efferverscence occurs	Very fast decomposition of hydrogen peroxide

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	ii)	Test the mixture with red and blue litmus paper.	Red litmus remained red and blue litmus remained blue	The mixture is neutral
4	i)	Add distilled water to cover the liver tissue, boil and 1cm3 of hydrogen peroxide.	No bubbles are given off/ No effervescence occurs	No decomposition of hydrogen peroxide
	ii)	Test the mixture with red and blue litmus paper.	Red litmus remained red and blue litmus remained blue	The solution is neutral

b) In which test tube was the reaction most vigorus?

In test tube 3

c) What is the nature of;

(i) Solution **X**?

Solution X is alkaline

(ii) Solution **Z**?

Solution Z is acidic

d) In what medium is the active substance in the liver tissue most active?

The active substance in the liver tissue is most active in a neutral medium

e) What is the effect of heat on the active substance in the liver?

Heat denatures the active substance in the liver, since there is no reaction as in test tube 4

f) Identify the active substance in the liver tissue; give a reason for your answer.

The active substance in the liver tissue is an enzyme called catalase

Reason: -It breaks down hydrogen peroxide to water and oxygen gas is liberated

-It is denatured by heat.

g) Which factors are being investigated in this experiment?

Effect of heat/temperature on the rate of enzyme activity

Effect of pH on enzyme activity

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CHAPTER TWO

2.0 DIFFUSION AND OSMOSIS

Diffusion is the movement of molecules from a region high concentration to a region of low concentration.

Osmosis is the movement of solvent (water) molecules from a solution of low solute concentration to the solution of high solute concentration across a semi-permeable membrane.

A student in this case will be provided with solutions of different concentrations and a specimen (majorly plant material/tissue). Experiments here may involve placing cut plant tissues e.g. stems, potato cylinders, pawpaw strips, and etc in solutions of different concentrations and measurements taken may be curvature of plant tissues, change in length of plant tissues.

2.1 CURVATURE OF PLANT TISSUE

Cut a plant stem for example *Commelina* or flower stalk of *Cana* lily, into 4 equal strips and place one in each solutions of different concentrations of sucrose, and leave to stand for some time. After this time, a student may be asked to observe the cut strips and feel them using his or her fingers.

In such an experiment, the stems curve and the degree of curvature is used to determine the extent of osmosis.

- Solutions whose concentration is near similar to that of the cell sap (**isotonic solutions**) have pieces of stem in them remaining strong and straight.
- Solutions whose concentrations are higher than those of the cell sap (**hypertonic solutions**) have strips in them curving inwards i.e. with epidermis outermost. They are also weak or softer due to flaccidity of their cells resulting from the movement of water from them by osmosis.
- Solutions whose concentrations are lower than those of the cell sap (**hypotonic solutions**) have strips in them curving outwards i.e. with epidermis inner most. Such strips are also found to be strong due to uptake of water by osmosis, which has cause the cell in them to be turgid.

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2.2 CHANGE IN LENGTH METHOD

Plant tissues e.g. potato cylinders of the same length and diameter (same volume) are placed in solutions of different concentrations e.g. sucrose and allowed to stand for sometime like 1 hour.
Then their lengths are measured and their nature recorded.
Results can be recorded as change in length (final length-initial length**)**

Or

Recorded as percentage (%) change in length

$$\text{Percentage (\%) change in length} = \frac{\text{New length} - \text{original length}}{\text{Original length}} \times 100\%$$

Results can be also be recorded in terms of ratios as initial length**:** final length.

The student may be asked to plot a graph of sucrose concentration against changes in length or percentage change in length.

2.3 EFFECT OF DILUTE SOLUTIONS (HYPOTONIC SOLUTIONS) ON PLANT TISSUES (CYLINDERS)

Observation

(i) Texture: plant tissues are hard and brittle

(ii) Size: plant tissues are longer than their original length

Explanation

Plant tissues (cylinders) placed in solutions more dilute than their sap take in water by osmosis and their cells become turgid.

2.4 EFFECT OF MORE CONCENTRATED SOLUTIONS (HYPERTONIC SOLUTIONS) ON PLANT TISSUES (CYLINDERS)

Observation

(i) Texture: tissues appear soft and flabby

(ii) Size: they are shorter than their length.

Explanation

Plant tissues place in stronger (more concentrated) solutions than their cell sap loses water by osmosis and their cells become flaccid.

2.5 EFFECT OF ISOTONIC SOLUTIONS ON PLANT TISSUES (CYLINDERS)

Observation

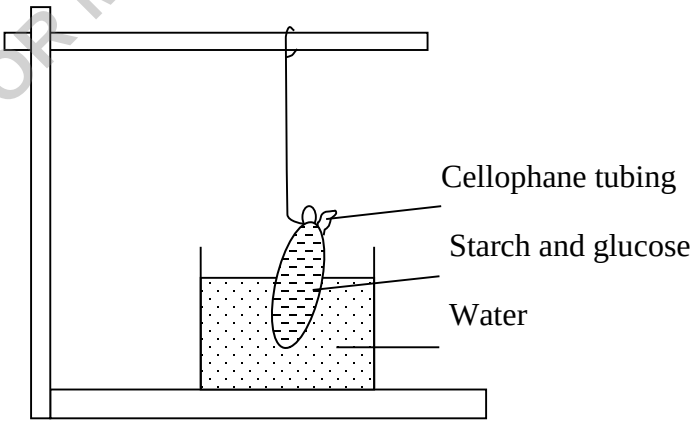
There is no change in length

Explanation

Plants tissues placed in solutions with the same concentration as that of their cell sap there is no net gain or loss of water by osmosis.

2.6 OSMOSIS VIA A SEMI-PERMEABLE MEMBRANE

Arrangement of apparatus



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A mixture of starch and glucose is placed inside a cellophane tubing/bag and tied very tightly. Then lowered into a beaker of distilled water and the experiment is allowed to stand for some time.

After this time water is tested for the presence of a reducing sugar using Benedict's solution, and for starch using iodine solution.

It is found that only glucose has diffused through (because it is made up of smaller molecules than starch) and the cellophane tubing acts as a semi permeable membrane.

Possible questions to be asked from such an experiment

a) What is the nature/structure of the cellophane bag/tubing

It is a semi permeable/ selectively permeable membrane

b) Which physiological or biological process is involved?

It is osmosis

c) What is the function of this process?

- It is for absorption and movement of water across the root
- For maintenance of turgidity and hence the support of stems of herbaceous plants
- It helps in the opening and closure of stomata and flowers.

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2.7 EXPERIMENTS

EXPERIMENT 18

You are provided with specimen **F** and seven sucrose solutions of 0.0M, 0.1M, 0.3M, 0.4M, 0.5M, and 0.6M.

Place 12cm^3 of each solution into separate test tubes and label them accordingly.

Using a sharp Knife/scalpel cut slices about 5mm thick. From the largest slices cut out a total of seven potato strips each about 5mm wide.
Trim all of them to a length of 50mm wide.
Immerse one potato strip in each solution and leave for at least 45 min. At the end of this period remeasure their lengths to the nearest millimeter.

(a) (i) Calculate the change in length and from this work out the percentage change in length. Record your results in a suitable table.

CHAPTER THREE

3.0 ANATOMY OF PLANTS AND ANIMALS

Anatomy is the bodily structure of organisms. A number from this section may be on topics such as insects, modified roots, stems, leaves, fruits, seeds and germination, flowers and inflorescence, birds, fish, feathers (birds) and bones (teeth).

A number of specimens may be provided and a student asked questions to identify each, classify, give similarities and differences, their habitats or habits, adaptations, draw the structure asked for, label it, stating the magnification and even describing it.

All answers to such questions should be based on use of external and observable features on these specimens whose details can be viewed using a hand lens.

3.1 DRAWING AND LABELING

- Always use a sharp HB pencil or clutch pencil or sketch pencil
- Draw what you see but not what you imagine.
- Draw large diagrams
- Don't shade or colour your diagram
- All drawings must be proportional to the whole diagram
- Always make sure that you include the magnification, preferably at the bottom right corner. The magnification of a drawing is the ratio of the size of the drawing to the size of the specimen.

$$\text{Magnification} = \frac{\text{Size of drawing}}{\text{Size of the specimen.}}$$

  It can be written as X2 or X3 or X4 or X0.5 e.t.c

  Parameters of size can be obtained using measurements of length or width of the specimen and drawing respectively.
- Each drawing must be given a title.
- All drawings should preferably be in an upright posture and labeled (Unless when not requested to do so).
- Label lines must touch actual part of the diagram being labeled. Straight label lines must be used, preferably horizontal lines on the right and left sides of the drawing. Do not use arrow heads or dotted lines.
- Label lines must not cross each other.
- Diagrams drawn must be drawn in the central area of the space provided and space for labeling should be put into special consideration.
- Avoid labeling in plural when labeling a single point or when only one line is used.
- Do not leave any gaps between the diagram and its individual component parts/ lines.
- Take special note of spellings when labelling. Some words have similar pronunciation but have different meanings such as a bud and bad, west and waste, axillary and auxiliary.

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3.1.1 VIEW AND SECTIONS:

A table showing drawing of views and sections

View/section	Description
T.S = transverse section(horizontal section or cross section)	This is cut in the plane at right angles to the longitudinal section i.e. across the fruit between the two scars
L.S =longitudinal section	Involves cutting across the specimen's best line of symmetry for example in fruits it is cut from the point of attachment of stalk to the scars.
Dorsal view	(Back) side as seen from the back of the organism
Lateral view	drawing the side view i.e left side and right side of the specimen
Anterior view	Side as seen from the front of the organism i.e it's upper parts
Posterior view	(Hind) side as seen from the rear or hind part. Posterior part is the rear part.
Interior part	Inner part
Exterior part	Outer part
Trunk	Part of the body of an organism, without appendages, such as limbs, the tail and the head on an animal, branches and roots on a tree
Caudal	Part of the tail (tail region)
Proximal end	Upper or top of a drawing or specimen
Distal end	lower end or bottom part of a drawing or specimen
Ventral	Abdominal side

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3.2 CLASSIFICATION:

Classification is the grouping of organisms according to their similar characteristics. Organisms with similar characteristics are placed in the same group while those with different characteristics are placed in another group. Similarly a group of organisms which are found in the same group, can be separated and placed in respective groups, depending on their different characteristics.

3.2.1 Dichotomous key:

Organisms can be identified using their characteristics by using a **dichotomous key** or **an identification key. Dichotomous key** is therefore a tool for identifying organisms basing on their characteristics.

3.2.2 The characters used in the dichotomous key:

- Characters should be permanent i.e. should not change with the environment.
- Characters should be visible and tangible.
- Avoid using characters involving size, age or colour because these characters may change with time.
- Use descriptive features but not groupings. E.g. if it is an insect, use features found in an insect like three main body parts, pair of wings e.t.c. instead of saying it is an insect.

- Don’t use action words but use features enabling that action to take place e.g. possesses sucking mouth parts instead write possess proboscis.
- Don’t use one character more than once.

3.2.2 How to construct a simple dichotomous key?

A dichotomous key can be constructed using a collection of specimens.

- First take a clear observation of each specimen and note the features on them that will allow you to separate them.
- Divide the specimen into two groups depending on contrasting features and continue dividing each group into two at each level until you remain with one spacemen.
- The members of each group may not necessarily be equal, one group can even have member.
- The easiest way of constructing a dichotomous key is by using a comparison table. This allows you to look at a given character throughout all the specimen and then see how they differ.
- After obtaining the contrasting characters in the table, then you may use a comparison tree or flow chart or spider key.

Note: A comparison tree is your rough work which must never appear in the space meant for the dichotomous key, the same applies to the table unless it has been asked.

Example 1:

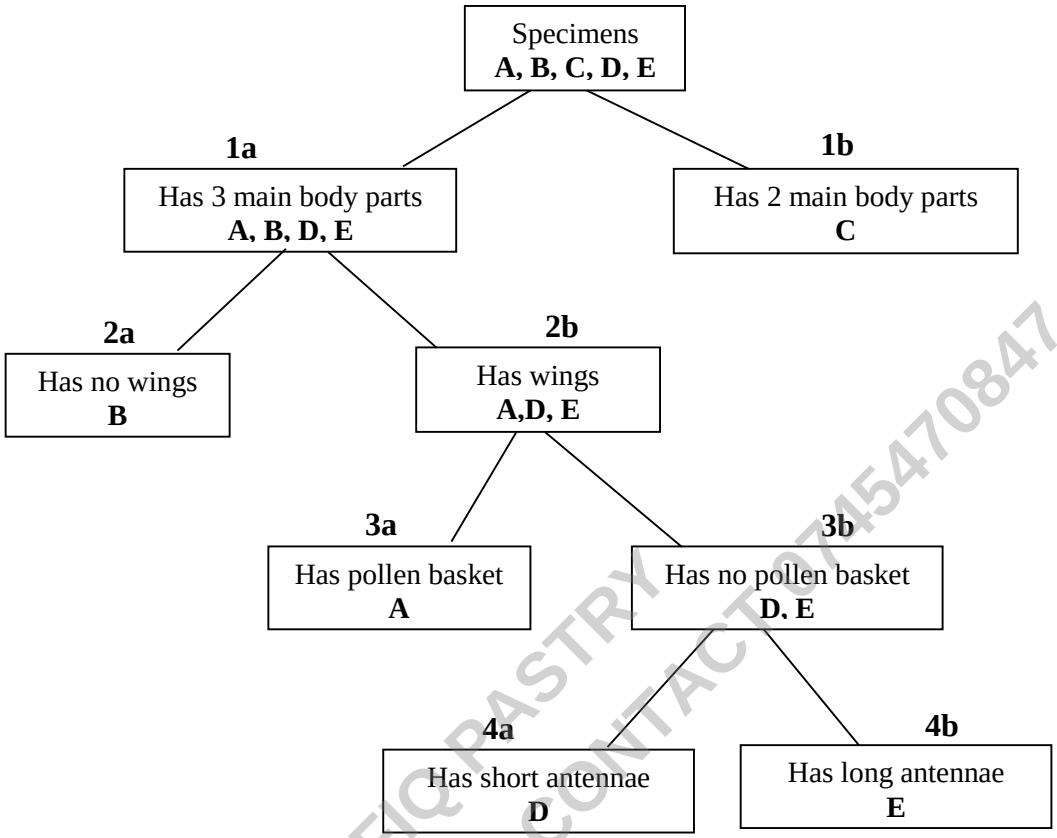
You are provided with the following specimens A (worker bee), B (soldier termite), C (tick), D (housefly) and E (cockroach). Study them and construct a dichotomous key.

Comparison table

Character	**A**	**B**	**C**	**D**	**E**
Main Body parts	3 body parts	3 body parts	2 body parts	3 body parts	3 body parts
Pairs of legs	3 pairs	3 pairs	4 pairs	3 pairs	3 pairs
Wings	Present	Absent	Absent	Present	Present
Pollen basket	Present	Absent	Absent	Absent	Absent
Antennae	Short antennae	Short antennae	No antennae	Short antennae	Long antennae

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Flow chart/comparison tree



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From the above flow chart the dichotomous can be constructed.

1 a) Has three main body parts……A, B, D, E (go to 2)
b) Has four main body parts……..C

2 a) Has no wings …… B
b) Ha wings ……..A, D, E (go to 3)

3 a) Has pollen basket ……A
b) Has no pollen basket ……D, E (go to 4)

4 a) Has short antennae ……...D
b) Has long antennae ……E

N.B The number of levels must be n-1 where n is the number of specimens. For example in the above dichotomous key, the number of specimen is 5. Therefore, the number of levels = 5-1 = 4

3.3 PLANT PARTS AND STRUCTURE

Plants are members of kingdom Plantae, which is divided into two divisions equivalent to phyla. These are spermatophyta and pteridophyta.

3.3.1 Flowering plants:

Flowering plants belong to division spermatophyte and class Angiospermae, which is divided into monocotyledonous and dicotyledonous plants.

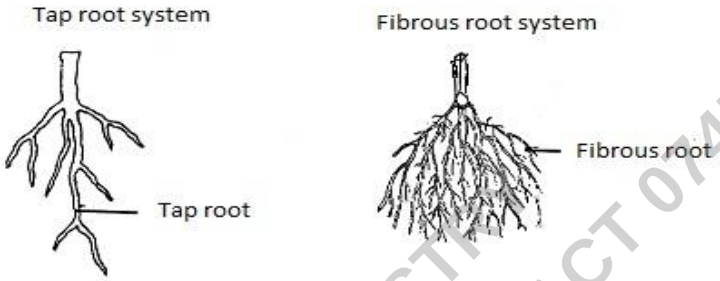
Flowering plants are divided into two main systems/parts:

(a) Shoot system: This is the part found normally above the ground. It consists of the following parts; stems, leaves, and flowers/buds.

(b) Root system: This is the descending part of the axis of the plant. There are main types of root system

(i) Tap root system: The main root of this system is called tap root and grows downwards into the soil from which other roots (lateral roots) branch from. This a characteristic of dicotyledonous plants e.g. beans, peas, soya beans and ground nuts.

(ii) Fibrous root system: This is a root system without a main root and all roots arise from the same point at the base of the stem. This a characteristic of monocotyledonous plants e.g. millet, maize, sorghum, rice, wheat e.t.c.



3.3.2 Modified roots:

These subdivided into modified tap roots and modified adventitious roots.

(a) Modified tap roots:

(i) Storage roots: These are thick fleshy and succulent roots, also called root tubers. They for modified for storage of food. They include cassava roots and carrots.

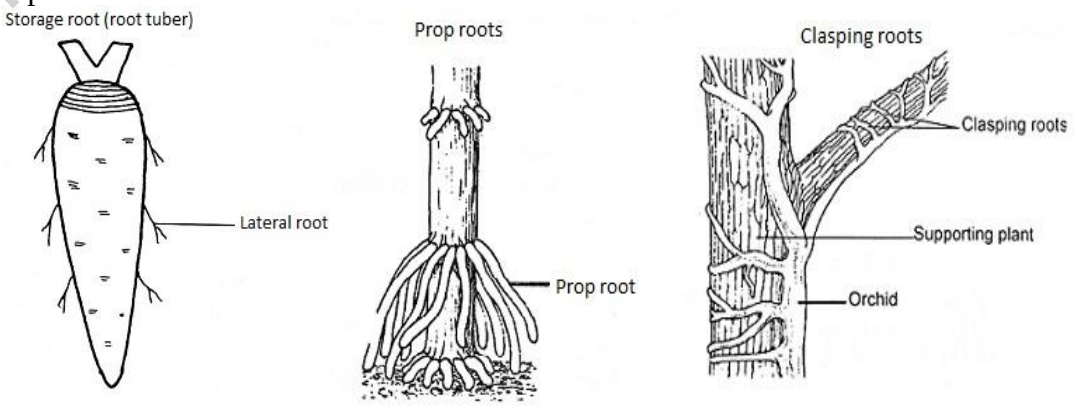
(b) Modified adventitious roots:

(i) Storage roots: These are modified adventitious that grow from the nodes of creeping plants. These roots are also known as root tubers. They are modified for food storage. Examples include; sweet potato roots.

(ii) Prop roots: These roots develop from nodes of a stem close to the soil surface. They provide extra support to the plant. They are found in maize, sugar cane, sorghum etc.

(iii) Clasping roots: These are roots growing from the nodes of climbing stems of some plants. They provide extra support to the plant.

(iv) Sucking roots: These are roots found growing on certain parasitic plants, for example, dodder plants. They absorb water, mineral salts and organic food compounds from the host plant.



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3.3.3 STEMS

A stem is the ascending portion of the plant axis that develops from the plumule of an embryo. They bear leaves, buds and flowers. Points on the stem were leaves or buds arise are called nodes and the regions between successive nodes are called internodes.

Most stems are found above the ground, however there are some stems that are found under the ground and these are referred to as underground stems.

Stems generally conduct materials like water, mineral salts, manufactured food and some manufacture and store food.

3.3.4 Modified stems:

Stems that are considered normal, are located above the ground are grow upright above the ground. Stems that are different from the normal ones are called modified stems and most of them are found under the ground. Modified stems have additional functions.

Modified stems include;

a) Rhizome

This is a horizontal thick underground stem having adventitious roots growing from the lower side of the nodes.

It has terminal buds which develop into aerial shoots and it bears buds in axils of scale leaves.

They store food for the plant and also acts as organs for vegetative propagation or reproduction.

For example, ginger, couch grass, spear grass e.t.c

b) Stem tuber

This is a short fleshy underground stem swollen to enable food storage.

It has scale leaves and axillary buds which form the eyes. For example, Irish potatoes.

They store food for the plant and also acts as organs for vegetative propagation or reproduction.

c) Bulb

This is a short conical shaped underground stem comprising of thick fleshy leaves arranged in concentric circles. Fleshy leaves store food for the plant and are protected by scale leaves.

Bulbs also contain buds used for vegetative propagation e.g Onions and garlic.

d) Corm

A corm is a swollen fleshy underground stem that grows in a vertical direction. It has a terminal bud lying at the top of the stem and has scale leaves arising from the nodes. E.g Cocoyam

They store food for the plant and also acts as organs for vegetative propagation or reproduction.

3.3.5 Other stems

a) Climbing stems

These are stems that grow clinging to a support by means of tendrils e.g in Passion fruits.

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b) Twinner / Twining stems

These are stems that grow ascending spirally around the support e.g. in bean plant, yams (Balugu).

c) Stolons

These are horizontally growing stems that have roots at the nodes and develop buds that grow into new plants. E.g Straw belly

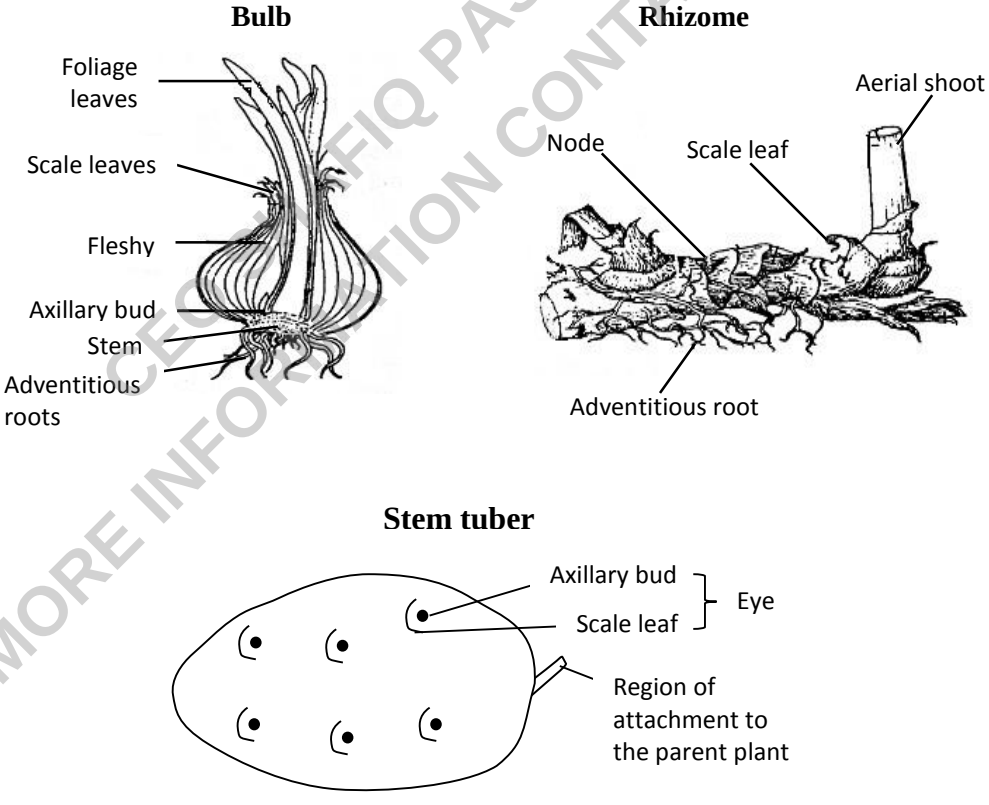
d) Runners

Runners are slender trailing stems lying flat on the ground with long internodes, and adventitious roots e.g Commelina sp (wondering jaw) and oxalis.

e) Suckers

A sucker is a creeping stem that grows obliquely upwards, directly giving rise to a leafy shoot. E.g Banana, pineapple, sisal plant e.t.c.

N.B: Runners, stolons, suckers are collectively called creepers or creeping stems



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3.3.6 LEAVES

This is a thin flattened structure which grows laterally from the node of the stem. It is where photosynthesis takes place.

A typical leaf is made up of three main parts namely

(i) **Leaf base**: This is the part that attaches the leaf to the stem.

(ii) **Leaf stalk** (**petiole**): This connects the leaf base to the leaf blade.
(iii) **Leaf blade** (**lamina**): This is the expanded and flattened portion of leaf.

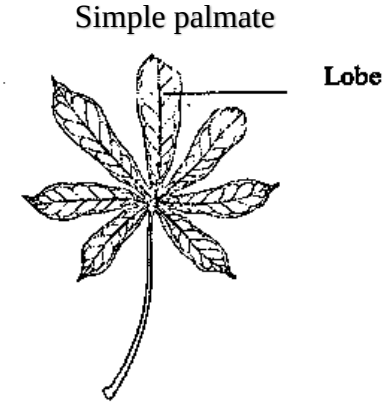
Leaf venation:
This is the arrangement of veins in the lamina of a leaf. There are two types of venation namely;

(i) **Reticulate venation**: Here the veins in the lamina branch while intersecting to form a net. These are found in dicotyledonous leaves.
(ii) **Parallel venation:** In this venation, the veins run side by side without branching. These are found in monocotyledonous leaves.

Leaf margin:
There are several types of leaf margins. These include;

(i) **Entire margin**: This is a margin without indentations of any kind e.g mango leaves, orange leave.
(ii) Serrated margin: This is a margin with indentations pointing toward the apex. E.g Hibiscus and lack jack leaves.
(iii) Crenate margin: This is a margin with round indentation. E.g Bryophylum.
(iv) Wavy margin: This is a margin with wavy blunt curves e.g *Solanum incanum*
(v) Lobed margin: This is a margin in which the lamina is deeply divided but do not separated completely. E.g cassava, sweet potato, passion pawpaw, e.t.c

Types of leaves:
There are two major types of leave. These include simple leaves and compound leaves.



A. Simple leaves:
Simple leaves are leaves with a single lamina which is not completely divided up into leaflets e.g. mango, Jack fruit, orange etc. Pawpaw and cassava have simple digitate/palmate leaves.

B. Compound leaves:
Compound leaves are leaves with a single lamina which is completely divided up into leaflets.

Types of compound leaves:

i) ***Compound pinnate:***
Here leaflets are arranged in two rows opposite one another along the midrib (rachis) e.g. cassia leaf.

ii) ***Compound bipinnate:***
Here leaflets are arranged opposite each other on the rachis and each leaflet further divides into smaller units called pinnules e.g. Jacaranda leaf.

iii) ***Compound trifoliate:***

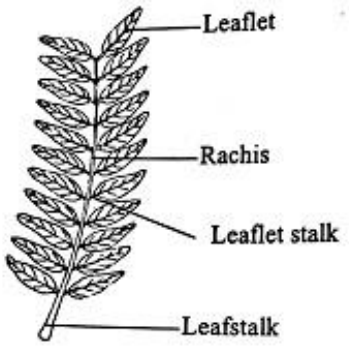
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These are compound leaves with only three leaflets e.g. Soya bean, beans, oxalis, straw berry, peas etc.

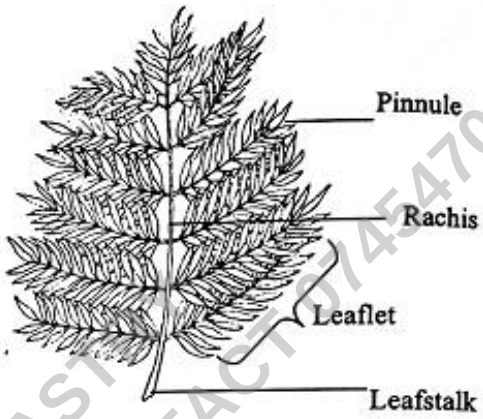
iv) ***Compound digitate or palmate:***

These are compound leaves with leaflets radiating out from the end of petiole like fingers of the hand e.g. Silk cotton and gynandropsis.

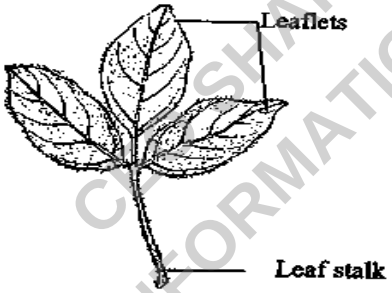
Compound pinnate



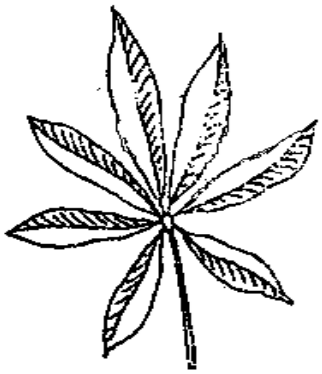
Compound bipinnate



Compound trifoliate



Compound palmate



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Note:

Lamina may also have the following features;

- May have hairs to trap moisture hence reducing transpiration.
- May be thick and succulent to store mainly water and food which is an adaptation for plants leaving in dry areas e.g. Sisal, cactus, and bryophylum.
- It may be rough which provides protection.
- It may have spines for protection against herbivores.

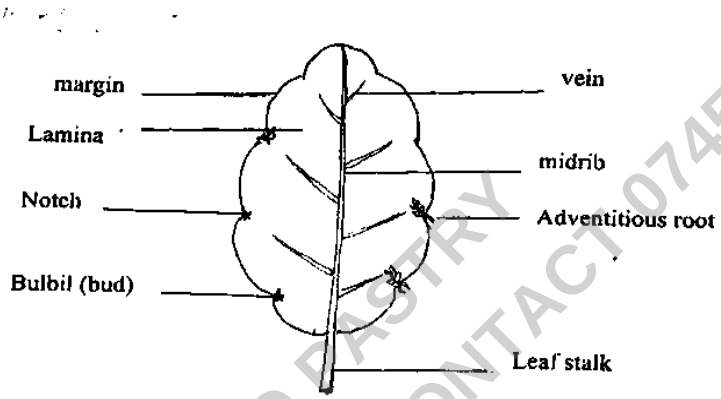
Leaf modifications:

Some leaves have been modified into specialized forms to perform other functions other than the usual photosynthesis, transpiration and gaseous exchange.

Functions of modified leaves include;

a) Some leaves store food and water for the plant. They are thick, succulent and swollen e.g. fleshy leaves of an onion, Bryophylum etc.
b) Some leaves are modified for vegetative propergation/reproduction. These have buds n the margin e.g. Bryophylum.
c) Some leaves are used for support and attachment on other objects. They have leaf tendrils at the apex or between the two leaflets e.g. peas.
d) Some leaves trap and digest insects e.g. *Nepenthes*. These leaves have leaf blades which arises as an outgrowth of leaf apex.
e) Some leaves provide protection e.g. pineapples, prickly pear, etc. These leaves have spines.

Bryophyllum leaf



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3.3.7 FLOWERS

A flower is part of a shoot specialized for sexual reproduction. The flower consists of several parts arranged in rings known as whorls. These whorls from outermost to innermost are namely;

a) ***Calyx (sepals):***

This is the outermost whorl of a flower. It is made up of sepals that are normally green in colour. However, in some flowers the sepals maybe brightly coloured, hence referred to as *petaloids*.

Sepals maybe either free (polysepalous) or fused (gamosepalous).

Calyx protect the inner whorls of a flower during the bud stage.

b) ***Corolla (petals):***

This is the second whorl of a flower and it is made up of petals which are normally brightly coloured to attract pollinators. The petals can either be free (polypetalous) or fused (gamopetalous).

In monocots, petals are small and dull coloured combined with sepals to form a ***perianth***.

c) ***Androecium (stamens):***

This is the third whorl of a flower made up of stamens. Stamens form the male part of the flower and each consists of a filament which supports a head called anther.

d) ***Gynoecium (pistil/carpels):***

This is made up of female reproductive parts called carpels. Each carpel is made up of ovary, style and stigma. A flower with one carpel is referred to as ***monocarpous***, that with several fused carpels is referred to as ***syncarpous*** and that with several unfused carpels are referred to as ***apocarpous.***

Other terms used to describe flowers:

(i) ***Compete flower*****:** This is a flower having all the four floral whorls.

(ii) ***Incomplete flower*****:** This is a flower lacking one or more of the four floral whorls.

(iii)*Bisexual flower*: This is a flower with both male and female parts i.e. it has pistil and stamens. These flowers are also called hermaphrodite flowers.

(iv)*Unisexual flower*: This is a flower with either male sexual parts or female sexual parts.

(v) ***Pistillate flower*****:** This is a female flower i.e. a flower with carpels and no stamens.

(vi)*Staminate flower*: This is a male flower i.e. a flower with stamens and no carpels.

(vii) ***Regular flower*****:** This is a flower which can be divided symmetrically along more than one plane.

(viii) ***Irregular flower*****:** This is a flower which can be divided in two similar halves along only one plane.

(ix) ***Superior ovary*****:** This is the ovary that arises above the rest of the floral parts.

(x) ***Inferior ovary*****:** This is the ovary that arises below the rest of the floral parts.

Pollination of flowers:

Flowers are generally pollinated by insects (insect pollinated flowers) and wind (wind pollinated flowers).

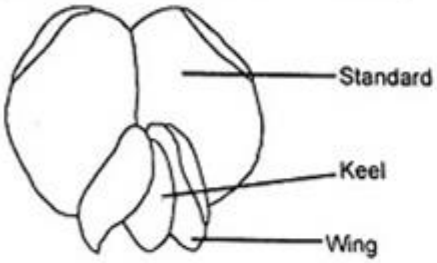
Characteristics of insect pollinated flowers.

- They have large and conspicuous petals to provide large landing area and to be easily seen by insects.
- They have brightly coloured petals to attract insects.
- They are scented to attract insects.
- They have sticky stigma on which pollen grains can stick.
- They produce sticky pollen grains so as to stick on the body of insects.
- They produce nectar that attracts insects.

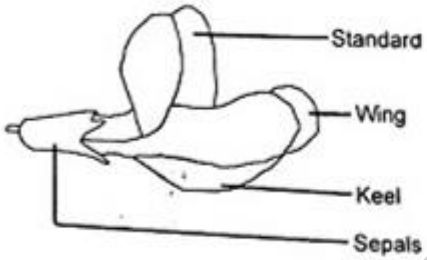
Examples of insect pollinated flowers include; crotalaria flower, hibiscus flower, morning glory, cassia flower e.t.c.

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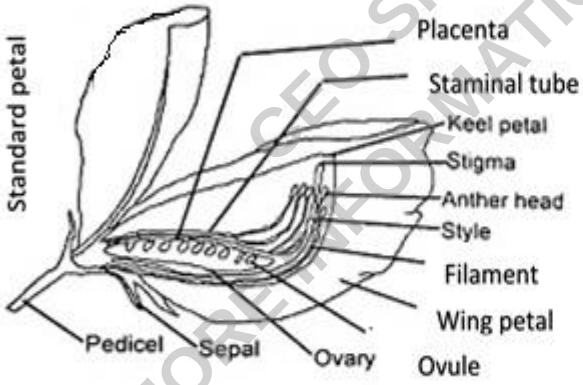
Anterior view of crotalaria flower



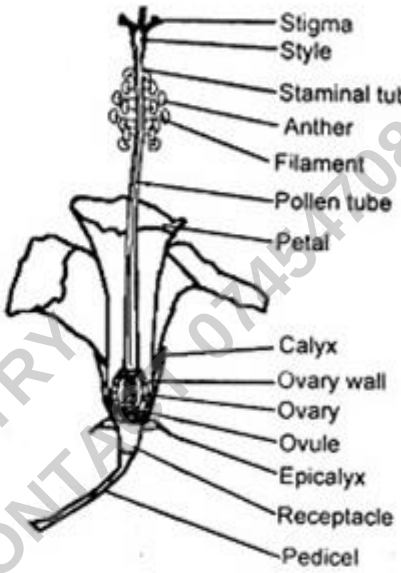
Lateral view of crotalaria flower



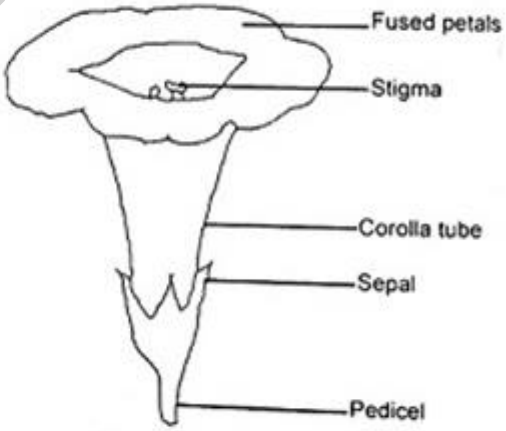
Longitudinal section of crotalaria flower



Longitudinal section of Hibiscus flower



External parts of morning glory flower



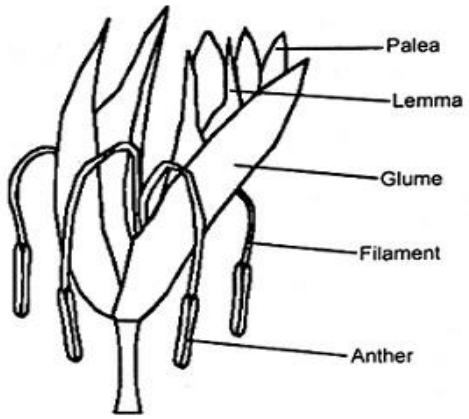
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Characteristics of wind pollinated flowers.

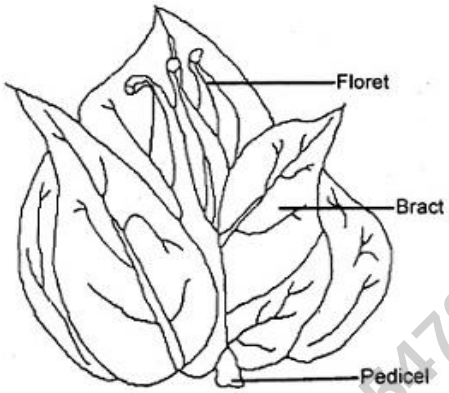
- They have feathery stigma which provide a large surface area for filtering pollen grains from the air.
- They have long styles for protruding the stigma outside.
- They have loosely attached anthers.
- They produce many dusty small pollen grains which are light in weight.
- They have long thin filaments for holding the anthers outside the flower.
- They have small dull coloured petals or perianth.

Examples of wind pollinated flowers include; grass flower and maize flower.

Drawing of a maize spikelet



External structure of Bougainvillea flower



3.3.8 FRUITS AND SEED:

A seed is a fertilized mature ovule. It has one scar (hilium) which is a spot where it was attached to the pod inside the fruit using the funicle.

A fruit is a mature fertilized ovary containing or more seeds. It has a fruit coat known as a pericarp. It has two scars, one where it was attached to the receptacle and the other of the remains of the style or stigma.

CLASSIFICATION OF FRUITS

Fruits are divided into three groups namely;

(i) Simple fruits:

These are fruits which are formed from one flower in which the pistil is either monocarpous or syncarpous.

(ii) Aggregate fruits:

These are fruits formed from one flower with gynoecium having several free carpels (apocarpous).

(iii) Multiple fruits:

These are formed from several flowers and the ovaries become fused after fertilization e.g Jack fruit, pineapple.

SIMPLE FRUITS

These are the majority of fruits and are divided into dry fruits and succulent fruits.
Dry fruits are divided into dry dehiscent and dry indehiscent fruits.

a) Dry indehiscent fruits:

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These are single seeded fruits with a pericarp which does not split to set free its seed. These fruits include achene, nut, caryopsis, cypsela and samara.

(i) Achene:

This is a small one seeded dry fruit with a tough leathery pericarp which does not split or dehisce. For example sun flower.

(ii) Nut:

This is a one seeded fruit with a very hard and tough pericarp e.g. cashew nut.

Note: Ground nuts and coconut fruits are **not** nuts.

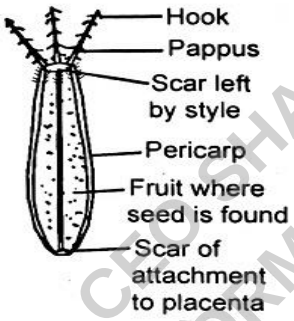
(iii) Caryopsis:

This is achene-like fruit in which the pericarp and testa have become fused together e.g. maize.

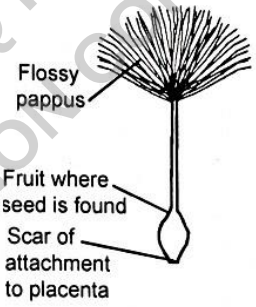
(iv) Cypsela:

This is a small hard one seeded fruit with a persisted calyx above the inferior ovary which forms a parachute of hairs known as ***pappus*** that facilitate wind dispersal by increasing the surface area that can be acted on by wind maximally. E.g tridax, black jack, Dandlion etc.

Bidens pilosa (Black jack)



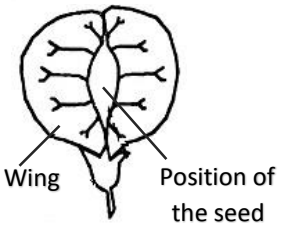
Dandilion:



(v) Samara:

This is an achene like fruit in which the pericarp grows flattened to form a wing which facilitates wind dispersal by increasing the surface area that can be acted on by wind maximally. E.g African rose wood.

Elm-Samara



b) Dry dehiscent fruits:

These are fruits where the pericarp on drying splits open (dehisces) to release seed. They split along lines of weakness referred as **sutures**.

These fruits are classified according to the number of splits (sutures) which occur in the pericarp. These include follicle, legume, capsule and schizocarp.

(i) Follicle:

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This is a dry many seeded fruit that splits longitudinally along one line of weakness e.g Sodom apple.

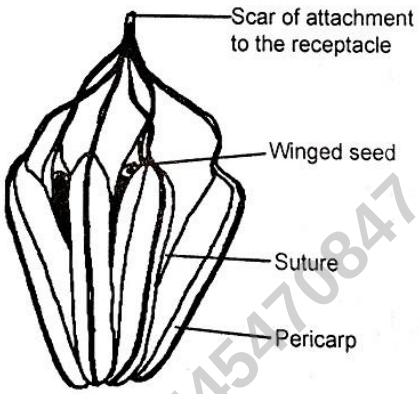
(ii) Legume:

This is a dry many seeded fruit that split longitudinally along two lines of weakness e.g. bean pod, pea pod, cassia pod etc.

(iii) Capsule:

This is a dry many seeded fruit that split longitudinally along more than two lines of weakness e.g Dutchman's pipe, castor oil, cotton, etc.

Diagram of dutchman's pipe-Capsule



(iv) Schizocarp:

This is many seeded fruit which on drying breaks up into several parts, each called a mericarp containing only one seed e.g. *desmodium.* They possess sticking hairs that attach to the body/fur of animals to facilitate animal dispersal.

SUCCULENT FRUITS:

These are fruits which become juicy as they ripen. Part of these fruits are fleshy and edible. The pericarp is divided into three layer i.e. epicarp, mesocarp and endocarp. Succulent fruits include drupe, berry and pome.

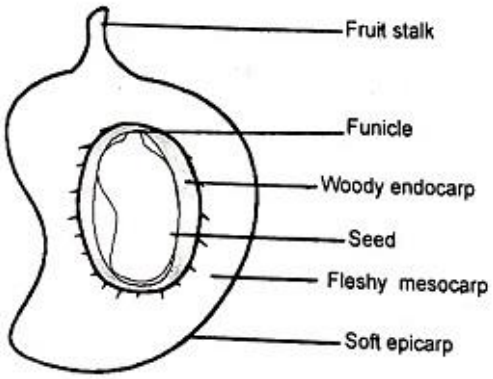
a) Drupe:

These are single seeded succulent fruits with a pericarp divided into three layers namely;

- ✓ ***Epicarp*** – an outer thin soft layer.
- ✓ ***Mesocarp –*** a middle fleshy edible layer.
- ✓ ***Enocarp –*** an inner hard and woody layer which forms the stone.

Example include avocado, mango fruit, coconut, etc.

Longitudinal section of a mango fruit:



Characteristics of drupes:

- Has one seed.
- Has fleshy mesocarp.
- The endocarp is hard and fibrious.
- Has only a single chamber (locule) where a seed is located.
- Seed is attached at the base of the fruit i.e. has basal placentation.

b) Berry:

This is a juicy fruit with many seeds e.g. tomato, orange, etc. The whole of its pericarp is soft.

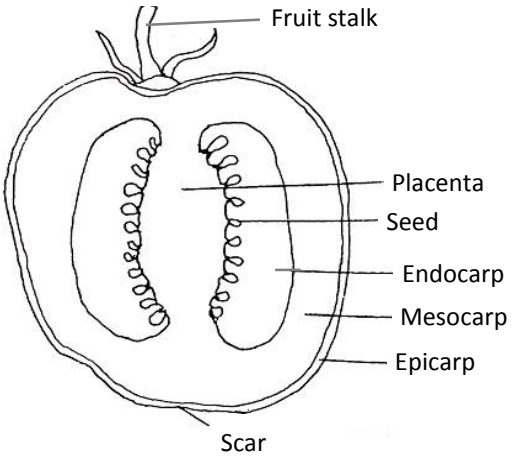
Characteristics of berries:

- Have many small seeds.

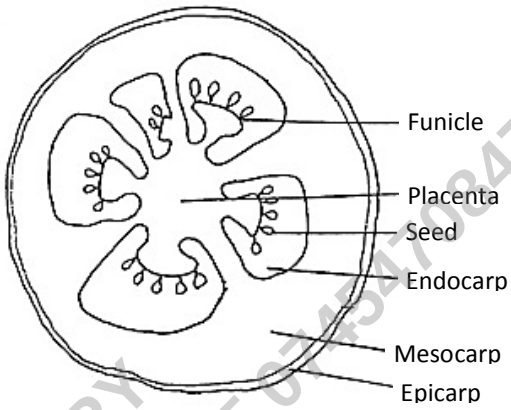
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- Have a spongy mesocarp.
- Endocarp id juicy, with several juicy hairs or sac in some fruits.
- Have several chambers (loculi).
- Seeds are attached around the placenta centrally (axile placentation).

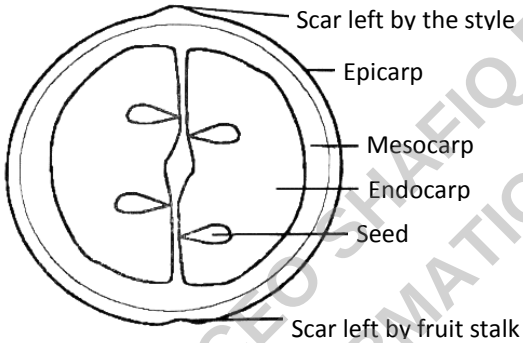
Longitudinal section of a tomato fruit:



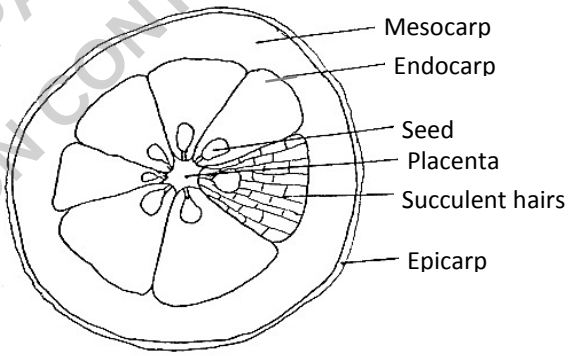
Cross section of a tomato fruit:



Longitudinal section of an orange fruit:



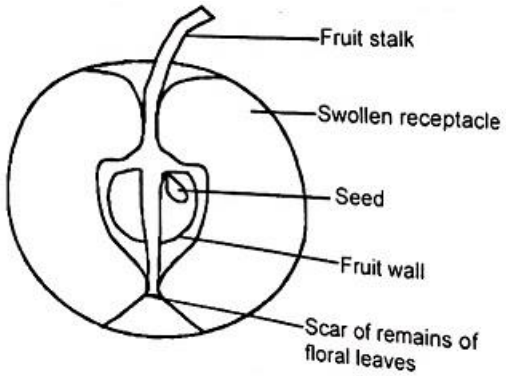
Cross section of an orange fruit:



c) Pome:

This consists of a fleshy pericarp whose receptacle forms part of the fruit e.g. apple.

Longitudinal section of an apple:



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3.3.9 PLACENTATION:

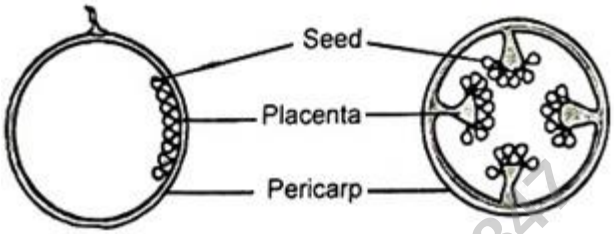
This is the arrangement of seeds and their attachment on the placenta.

There are several types of placentation and these include;

a) Axile placentation:

Seeds arrange themselves radially/centrally around the placenta which is centrally located e.g. oranges, tomatoes, lemon and banana.

Diagram of Pareital placentation of passion fruit:

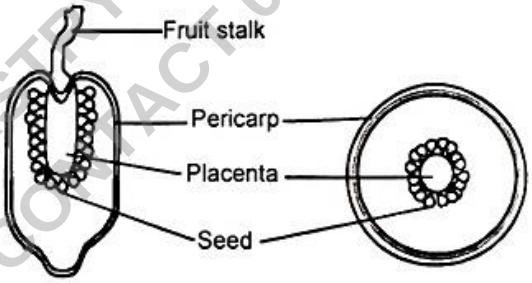


b) Marginal placentation:
Seeds are arranged along the placenta located on the margin of the pod of fruits like legumes e.g bean pod, pea pod, or follicles for example Sodom apple.

c) Basal placentation:
Here seeds are attached at the base of the pericarp e.g. mango, avocado, etc.

d) Parietal placentation:
There the seed are attached to the inner walls of the pericarp (endocarp) and appear as ridges e.g. pawpaw, passion fruit, cucumber, etc.

Diagram of free central placentation of green paper:



e) Free central placentation:
Here the placenta projects freely in the center of the locule (chamber) from where the seeds are attached all around the placenta e.g. green paper.

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3.3.10 SEEDS AND FRUIT DISPERSAL:
This is the scattering of fruits and seeds from their parent plants.

Importance of seed and fruit dispersal:

- It reduces overcrowding among plants of the same species.
- It reduces competition for resources.
- It allows colonization of new areas.
- It minimizes the spread of diseases.
- It enhances chances of survival and continuity of plant species.

Types / modes of dispersal:
There are four types of dispersal namely; animal dispersal, self-dispersal (explosive dispersal), wind dispersal and water dispersal.

a) *Animal dispersal:*
Fruits dispersed by animals have the following characteristics;

- Some are large and brightly coloured to attract animals.
- Some fruits when are scented to lure animals.

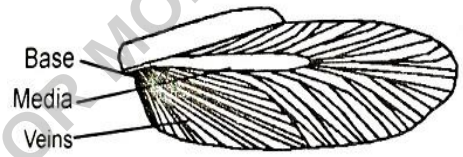
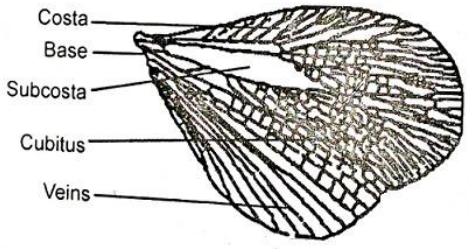
Features on the thorax

- Has three segments i.e prothorax, mesothorax and metathorax.
- Each segment bears a pair of legs. The hind leg have spines for protection, they are long and muscular and Z-shaped to provide enough lift force. The legs possess claws to grip on objects, glandular pads for moving on smooth surfaces. Legs are jointed for swift movement. The other legs are short for swift movement.
- The mosothorax and metathorax each possess a pair of wings.
- The outer wing is hard, narrow and oval in shape whose main function is to protect the inner wings used for flying.
- The inner wing is membranous and flexible supported by a system of veins. It is broad and triangular in shape providing a large surface area.
- Have two pairs of spiracle.

Features on the abdomen

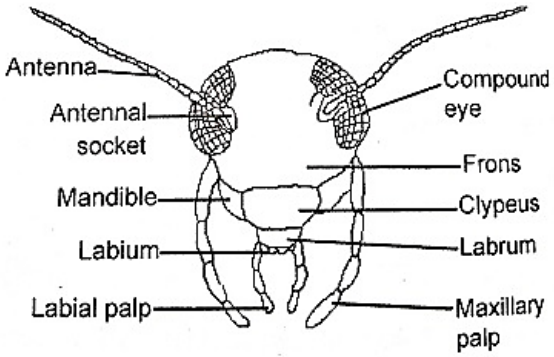
- Has ten segments when which are dorso-ventrally flattened.
- Each segment bears a pair of spiracles.
- The last abdominal segment has a pair of jointed sensory structures called cerci.
- It is flat and broad.
- The ninth segment in males possess a pair of styles which are reproductive structures.
- In the females the seventh segment possess the ovipositor used for holding and depositing the egg case.

Differences between fore wings and hind wings

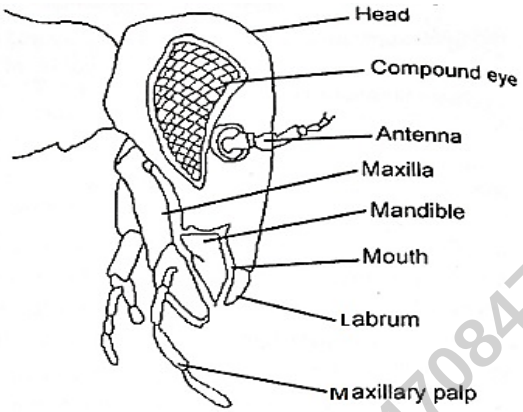
Fore wings	**Hind wings**
Are brown in colour and opaque	Are colourless and transparent
Are thick and stiff	Are thin, membranous and delicate
Are narrow shaped	Are broad shaped
Cover and protect hind wings not when in use	Used for flight
	

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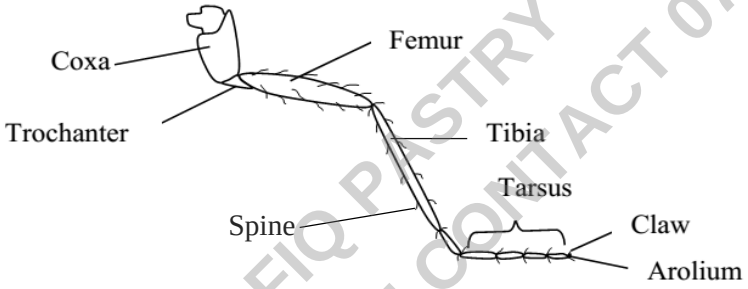
Anterior view of the head of a cockroach



Lateral view of the head of a cockroach



Drawing of hind leg of a cockroach



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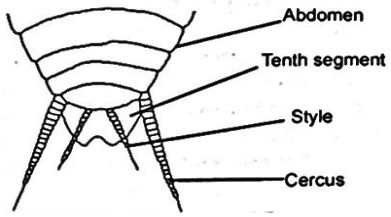
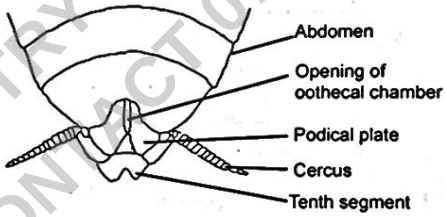
Adaptations of a cockroach to its habits and habitant

- They may be winged or wingless. Wings enable them to fly away quickly from enemies.
- They possess spines on their legs for defense.
- They have long jointed legs that they can use to run very rapidly from any potential danger.
- They are generally flat and oval in outline to enable them hide in very narrow crevices.
- They are brown or dark coloured allowing them not to be seen in darkness since they are largely nocturnal.
- They have a soft, slippery outer skin called cuticle which makes hem difficult to catch.
- They have long, highly sensitive antennae with sensory bristles which enable them to detect tiny amounts of food and moisture even in darkness.
- They have cerci which can sense minute air movements to enable a cockroach to rapidly detect and flee from potential danger.
- They have oily cuticle which protects them from dehydration.

Identification of a cockroach's sex

In males, there is a pair of slender **styles** that are used to hold and manipulate the female during copulation.

In females, there is a pair of boat shaped structures called the **podical plates** used for holding eggs.

Differences between a male and female cockroach:	
Male	**Female**
1. Has a narrow abdomen	Has a broader abdomen
2. Lack ootheca	Has ootheca which develops after fertilization.
3. Has rod-shaped structures called styles on the 9th abdominal segments.	No styles on the 9th abdominal segment.
4. No podical plates.	Has podical plate for carrying eggs.
	

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2. Housefly

Order: Diptera

Reason: Has a pair of wings.

Habitat: Dirty/filthy moist places/decaying organic matter.

Features on the head.

- Has two compound eyes
- Has three simple eyes called ocelli.
- Has a proboscis
- One pair of short antennae.
- Has a pair of short maxillary pulp.

Features on the thorax

- Has three segments.
- Each segment is hairy and each possess a pair of hairy legs.
- Mesothorax has a pair of membranous wings.
- Metathorax has a pair of haltares/balancers.
- Has two pairs of spiracles.

Adaptations of a housefly to its habits and habitant

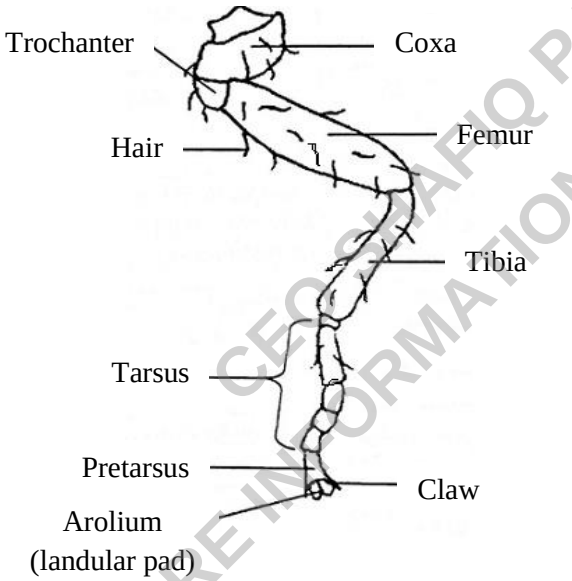
- Has hairy antennae for increased sensitivity.
- Membranous wings for flight.

- Tubular expanded proboscis for sucking liquid food and depositing of saliva.
- Large compound eyes for wide field of view.
- Curved/sharp/pointed claws for movement on rough surfaces.
- Sticky arolium for movement for moving on smooth surfaces.
- A pair of halteres for balancing during flight.
- Segmented body for flexibility during movement.
- Body covered with exoskeleton for protection against water loss/desiccation/mechanical damage.
- Has glandular pads on legs for moving on smooth surfaces.
- Has jointed limbs for flexibility during movement.

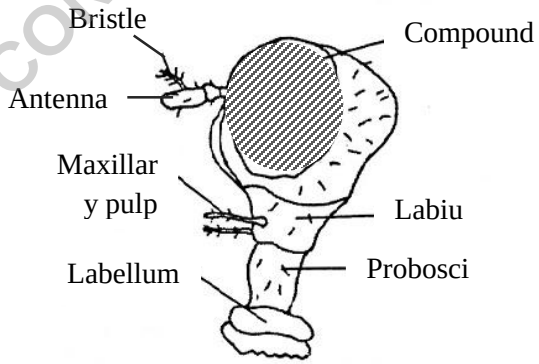
Adaptation of a housefly to its function as a vector

- Has hairy body for carrying germs to other areas.
- Has tubular proboscis through which germs contained in saliva are passed onto food
- Has wings for swift movement leading to easy transfer of the germs.

A drawing of a hind leg of a house fly



Drawing of lateral view of the head of a house fly:



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3. **Grass hopper**
   Order: Orthoptera
   Reason: Has straight wings.
   Habitat: Grass

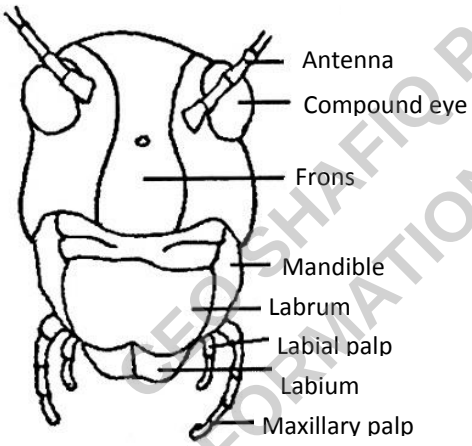
Features on the head

- Has a simple eye.
- Has a pair of short antennae.
- One pair of compound eyes.
- Have powerful mouthparts with strong toothed mandibles.

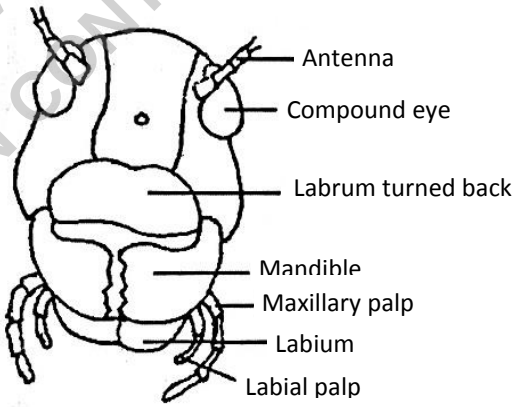
Adaptation of a grass hopper to its life in the habitat.

- Have a pair of antennae for increased sensitivity.
- Hard sharp mandibles for cutting solid food and defence.
- Curved/sharp/pointed claws for movement on rough surfaces.
- Sticky arolium for moving on smooth surface.
- Segmented body for flexibility during movement.
- Jointed maxillary palps to hold and push food into the mouth.
- Have dull coloured body to camouflage and avoid predators.
- Eyes are large and dorso-laterally positioned to see food in a wide area.
- Outer wings are hard to protect inner wings.
- Inner wings are wider for generating a big lift force in flight.
- Wings are veined for support in flight.
- Maxilla and labium are joined for flexibility.
- Jointed legs for flexibility during locomotion.

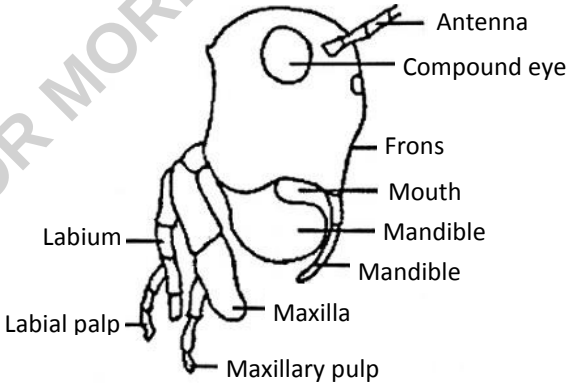
Drawing of anterior view of grasshopper head (All parts intact)



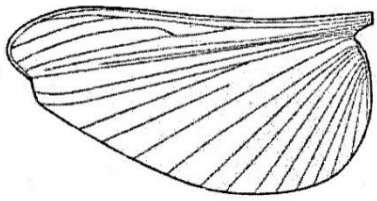
Drawing of anterior view of grasshopper head (Labrum lifted to expose mandibles)



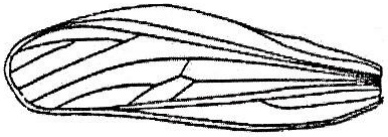
Drawing of lateral view of grasshopper head



Drawing of hind (inner) wing:

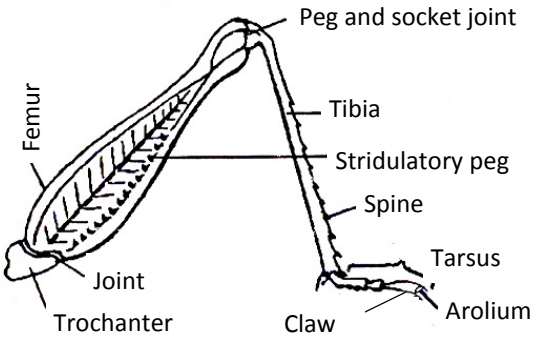


Drawing of fore (outer) wing:



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Drawing of hind leg of Grasshopper



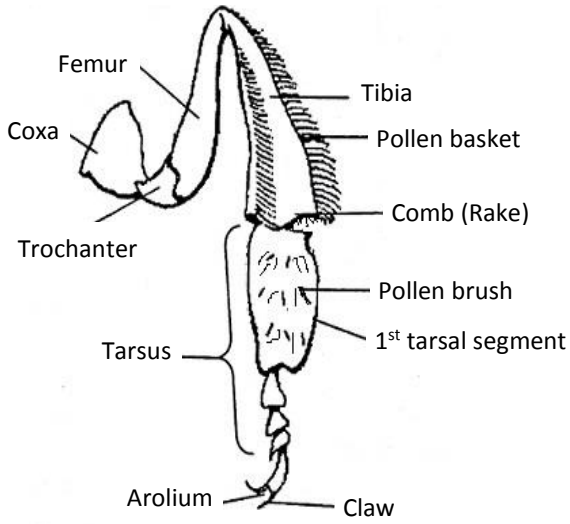
4. **Worker Bee**
   Order: hymenoptera
   Reason: Has membranous wings.
   Habitat: Bee hive

   Adaptations for successful survival of bees in their habitat
   - Possession of pollen baskets on limbs for carrying pollen grains.
   - Possession of pollen comb on their limbs for cleaning pollen off the body.
   - Possession of prongs on their limbs for removing the pollen from the pollen basket.
   - Hairy body for carrying the pollen grains.
   - Blunt mandibles for moulding wax and pollen grains.
   - Membranous wings for flight.
   - Spoon like tongue for lapping/sucking.
   - Segmented body for flexibility during movement.
   - Tubular expanded proboscis for sucking liquid food.
   - Large compound eyes for wide field of view.

   Structural adaptation of the worker bee to its function as a pollinator
   - Has hairy body to trap pollen grains.
   - Have membranous wings for swift movement leading to fast transfer of pollen.
   - Possession of pollen baskets on their limbs for carrying pollen grains.
   - Possession of prongs on their limbs for removing the pollen from the pollen basket.

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Drawing of the hind limb of the worker bee:



5. Termite

Order: Isoptera

Habitat: Ant hill.

Adaptations to the habitat

- Hairy antennae for increased sensitivity.
- Hard mandibles for cutting solid food and defence.
- Curved/sharp/pointed claws for movement on rough surfaces.
- Sticky arolium for moving on smooth surfaces.
- Segmented body for flexibility during movement.
- Body covered with exo-skeleton for protection against water loss/desiccation/mechanical damage.

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(f) Using a hand lens, study the head region including mouth parts of specimen G, draw and label the interior view. State your magnification.

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3.7 FISH

The ways in which the fish is adapted to live in water.

- Has stream lined body which enables it to move easily in water.
- Has scales which overlap backwards and the slippery skin, this reduces friction.
- Has fins which enable it to move in water and stop it from rolling.
- Has gills for breathing in water.
- Has lateral line under the skin containing sensory cells which enable it to sense vibrations in water.
- Has a wide mouth to allow water and food to enter easily.
- Has scales that provide protection.
- Has silvery colour below and dark colour above for camouflage.

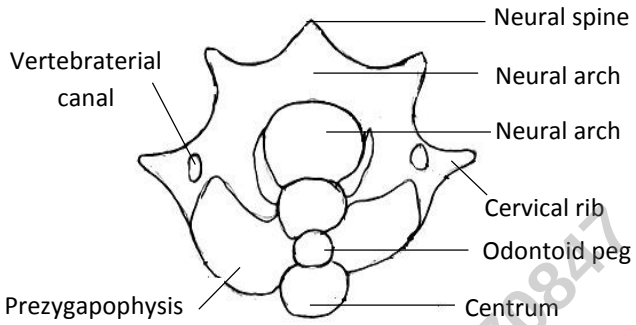
1. Thoracic vertebra

These are found in the thoracic region. There are 12 thoracic vertebrae in the humans and rabbits.

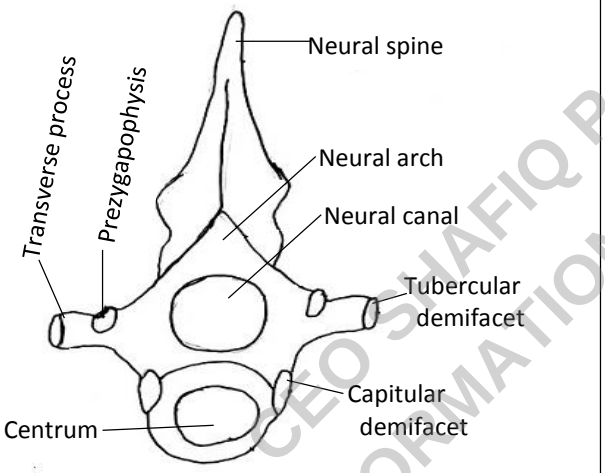
Characteristics of thoracic vertebra

i) Has a long neural spine.
ii) Has a large centrum.
iii) Possess a large neural canal
iv) Has a large neural arch.
v) Has tubercular facet on transverse process.
vi) Has short transverse processes.
vii) Has a pair of articular facets.

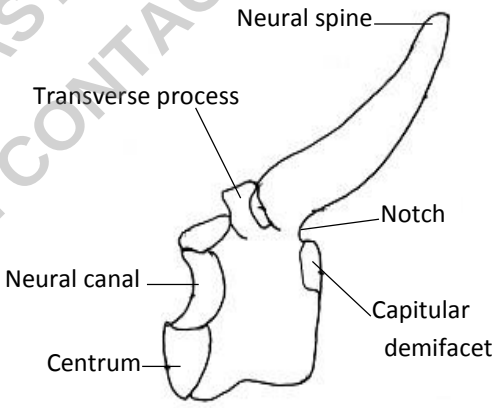
Drawing of anterior view of axis vertebra



Drawing of anterior view of thoracic vertebra:



Drawing of lateral view of thoracic vertebra:



2. Lumbar vertebra

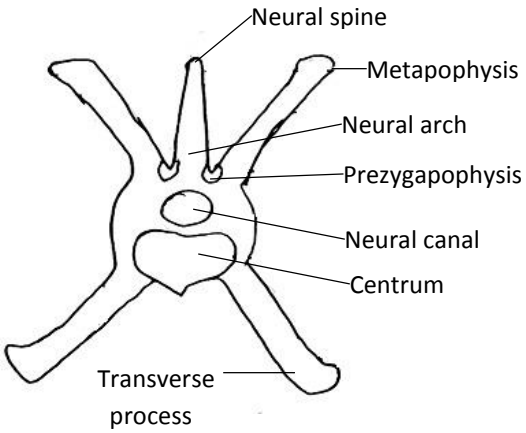
These are found in the abdominal region. There are 5 lumbar vertebrae in man and seven in rabbit. They provide the only support for the trunk in the abdominal region.

Characteristics of lumbar vertebra

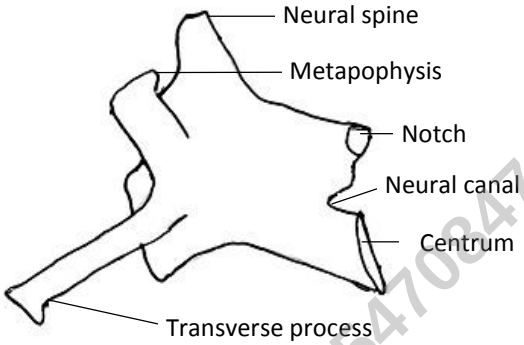
i) Has long transverse processes
ii) Has extra processes called metapophysis.
iii) Has a large centrum than cervical and thoracic vertebra.
iv) Has a broad neural spine.
v) Has a small neural canal.

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Drawing of anterior view of lumbar vertebra:



Drawing of lateral view of lumber vertebra:



Differences between thoracic and lumbar vertebrae.

Thoracic vertebra	**Lumbar vertebra**
Has long neural spine	Has short neural canal
Has small centrum	Has large centrum
Has short transverse process	Has long transverse process
Has fewer projections	Has many projections
Has a wide neural spine	Has narrow neural spines
Has no anapophysis	Has anapophysis
Has demifacets	Has no demifacets
Has a narrow neural arch	Has wider neural arch

Similarities between thoracic and lumbar vertebrae.

- Both have neural canal.
- Both have neural spine.
- Both have transverse process.
- Both have arches, facets and centrum.

3. Cervical vertebrae

This is found in the neck region. There seven cervical vertebrae in man.

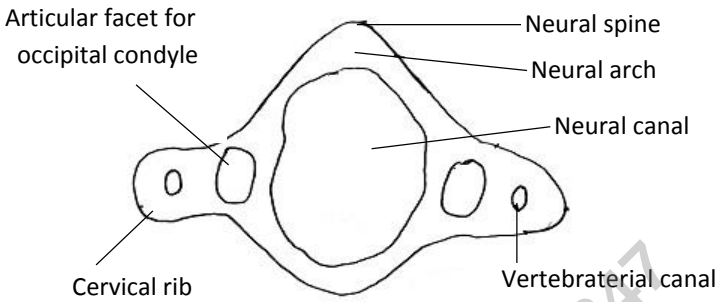
The first two cervical vertebrae, called atlas and axis respectively differ from the rest. They are respectively shaped to allow the skull move freely on the vertebral column.

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(a) Atlas

This is the first cervical vertebra

Drawing of anterior view of atlas vertebra



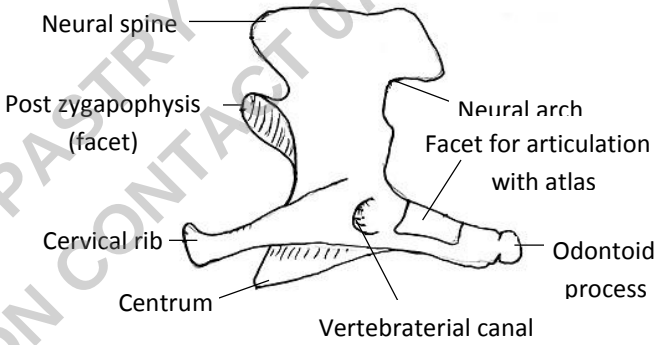
Characteristics of atlas

(i) Has very large neural canal.
(ii) It has no centrum.
(iii) Has flat broad transverse process for muscle.
(iv) Has two large facets.
(v) Has a small ridged neural spine.

(b) Axis vertebra

This is the second vertebra.

Drawing of lateral view of axis vertebra:



Characteristics of axis

(i) Has a relatively smaller neural canal than atlas.
(ii) Has a large flat centrum.
(iii) Has smaller transverse processes.
(iv) It has two facets.

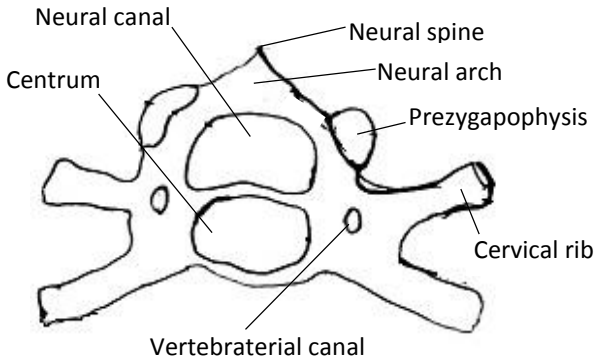
Differences between the atlas and axis vertebrae.

Atlas	Axis
Has no centrum	Has centrum
Has no neural spine	Has neural spine..
Has odontoid fossa	Has odontoid peg.
Has prominent process	Has no prominent process.

Similarities between the atlas and axis vertebrae

i) Both have transverse processes.
ii) Both have vertebrateral canals.

Drawing of anterior view of cervical vertebra



(c) Cervical vertebra

Characteristics of cervical vertebra

i) It has a small neural spine.
ii) Has a large neural canal.
iii) Has a small centrum.
iv) Has transverse processes that form two parts called cervical ribs.
v) Has vertebrateral canals

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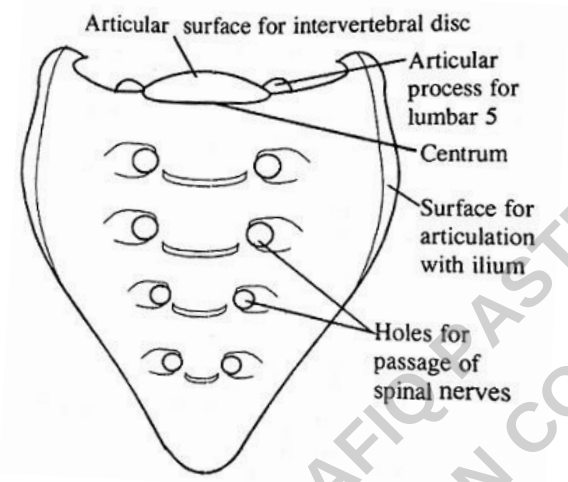
4. **Sacral vertebra**

These are located in the lower abdomen. They are five in man and four in rabbit. In adult mammals they are fused together to form a rigid body structure called sacrum.

Characteristics of sacral vertebra

(i) Each vertebra has a large centrum
(ii) Has a narrow neural canal.
(iii) Has reduced neural spine.
(iv) First sacral canal has large and wing like transverse processes.

Anterior view of sacral vertebra:



5. **Caudal vertebra**

These are found in the lower abdomen. There are four vertebrae in man which are fused together to form coccyx which is functionless. In rabbits, there are sixteen.

Characteristics of caudal vertebra

(i) Have no neural arch.
(ii) Have no transverse processes.
(iii) Have no neural canal and spine
(iv) Entire bone consists of centrum only.

Functions of parts of vertebra

Part	Function
Neural spine	For attachment of muscles
Transverse process	For attachment of muscles and articulation with ribs.
Centrum	For articulation with other vertebrae
Neural canal	For protecting spinal cord
Neural arch	For passage and protection of spinal cord
Vertebraterial canal	For passage of blood vessels of the neck
Facets	For articulation with other vertebrae and ribs in the thoracic vertebrae.

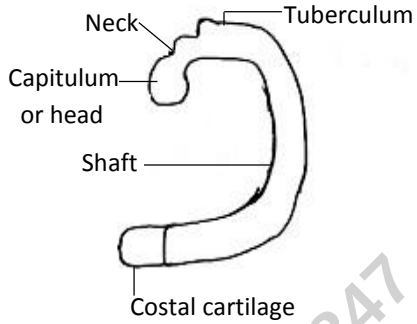
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3.8.2 RIB

It is long curved bone which form part of the rib cage.
Characteristcs of a rib

(i) It has costal cartilage.
(ii) It has a neck
(iii) It has a head
(iv) It has shaft which is thin and flat
(v) It has tuberculum.

Drawing of the rib

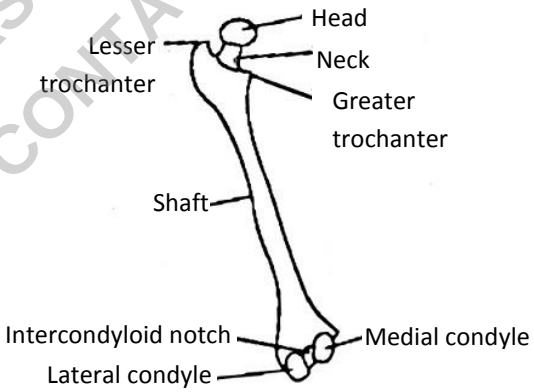


3.8.3 APPENDICULAR SKELETON

This consists of four limbs attached to two bony girdles namely pectoral (shoulder girdle) and pelvic girdle.

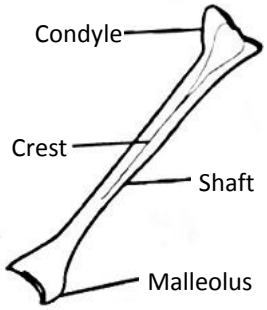
1. **Femur**
   It has a distinct head, which fits into the acetabulum of pelvic girdle to form a hip joint.
   It has trochanter near the head for which provide surface for muscle attachment.
   It has two knobs at the distal end.

Drawing of left Femur (posterior view):



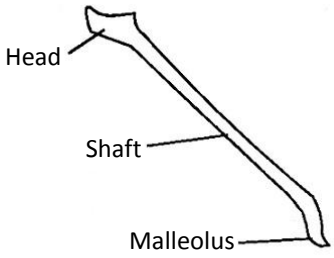
2. **Tibia**
   This is the inner and larger bone in the lower leg, extending from the knee to the ankle.
   It has two grooves into which condyles of the femur fits.

Drawing of left Tibia:



3. **Fibula**
   This is the outer and narrower bones in the human lower leg between the knee and ankle.
   It has unfused proximal end.
4. **Humerus**
   The upper (proximal) part of humerus ends in a distinct head which fits in glenoid cavity of scapula.
5. **Ulna**
6. **Radius**

Drawing of left Fibula:

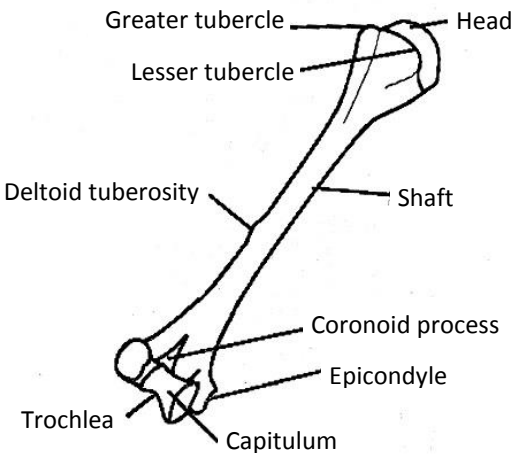


Drawing of Radius:

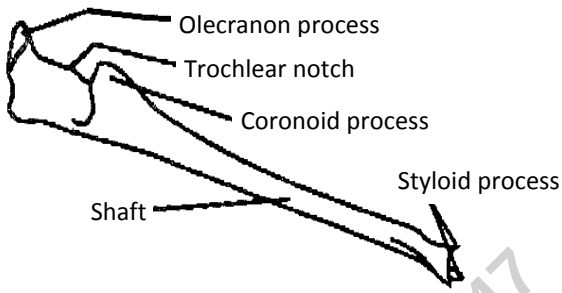


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Drawing of Humerus:



Drawing of Ulna:

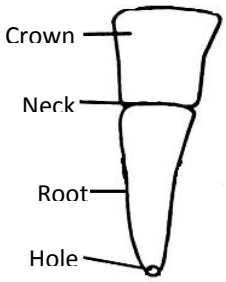


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3.8.4 TEETH

There are majorly four type of teeth i.e incisors, canines, premolars and molars.

Drawing of Incisor:



1. Incisor

These are front teeth in both upper and lower jaws of humans.

Characteristics of incisor

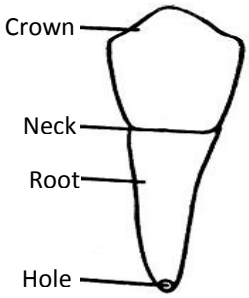
(i) It has a crown which is chisel-like in shape with sharp ridged edges.

(ii) It has one root.

Use

It is used for cutting and biting foo into smaller pieces. It some mammals like cats they are also used for holding prey.

Drawing of Canine:



2. Canine

These are found next to the incisors. They are usually larger than incisors in size.

Characteristics of canine

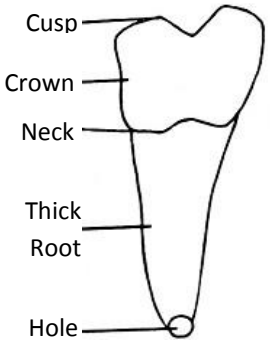
(i) Has a conical and sharp pointed crown.

(ii) It has one root.

Use

It is used for tearing food especially fresh.

Drawing of Premolar:



3. Premolar

It is found behind canine.

Characteristics of premolar

(i) It has a flat crown with a broad surface for grinding of food.

(ii) It has two cusps which enhance grinding of food.

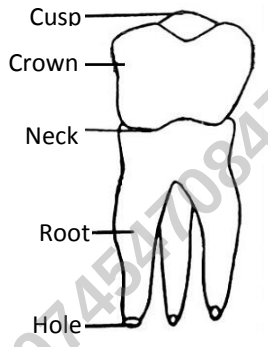
Use

It is used for grinding and chewing food.

4. Molar

It is found in the back of the mouth behind the premolar.

Drawing of Molar:



Characteristics of molar

(i) It has a wide crown top to provide large surface area for grinding and crushing of food.

(ii) It has four cusps to allow grinding and crushing of food.

(iii) It has two or three roots for fixing the tooth firmly in jaw to allow grinding of food.

Use

It is used for grinding and crushing of food.

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EXPERIMENT 51

You are provided with specimens **W**, **X** and **Y.** Study them carefully and answer questions that follow.

(a) Identify the specimens

W……

X……

Y……

(b) (i) Give the characteristics of specimen **W**

……

……

(ii) Give 5 differences between **W** and **X**

……

……

……

……

……

3.9 FEATHERS

These are structures that cover the entire body structure of the birds.

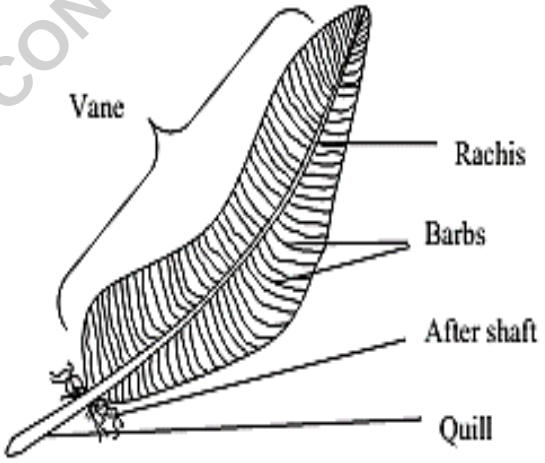
Function of feathers to birds.

(i) They protect the skin from abrasion, rain and direct sunrays.
(ii) They assist in maintenance of body temperature or regulating body temperature by insulating against heat loss.
(iii) They are used in flight
(iv) They are for camouflage to avoid predators.
(v) They are for courtship. Feathers are used to attract the opposite sex in birds.
(vi) They are for sensation, i.e. they collect sound waves

STRUCTURE OF A TYPICAL FEATHER

A typical feather consists of the following parts:-

1. **The shaft**. This is the framework or axis of the feather. The shaft is divided into the quill and rachis.
2. **The quill**. This is a cylindrical structure pointed at the end. It has two openings at its ends called inferior umbilicus and superior umbilicus.
3. **The rachis**. This extends from the superior umbilicus of the quill. It provides attachment of the vanes.
4. **The vane**. This is a feathery flat blade attached to the rachis. It consists of structures called barbs. The barbs have smaller parallel projections called barbules.
   The barbules strengthen the feather by making interlocking connections with each other.



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3.9.1 TYPES OF FEATHERS

1. **Quill feather/contour feather.**

Location: These are found on the tail and wing.

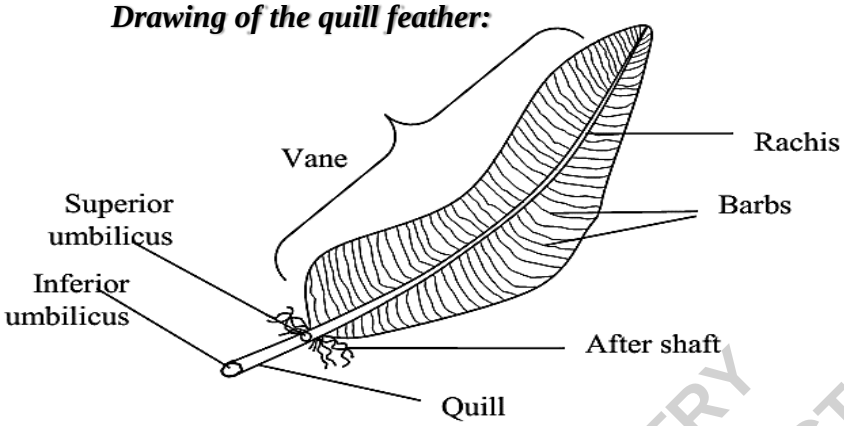
Characteristics

- They have a hard strong hollow quill.
- They have a large vane.
- They have a small after shaft.
- Their vanes consist of barbs and interlocking barbules.
- They have two holes, i.e. the inferior umbilicus and superior umbilicus.
- They have a large and long shaft.

Functions of the quill feathers

(i) They are used in flight.
(ii) They protect the skin of the bird.
(iii) They insulate the bird's body against heat loss.
(iv) They provide an air proof surface effective at breaking against air during flight.

Drawing of the quill feather:



Adaptations to flight

- Has a hollow quill, making it light, thus reducing weight.
- Has a broad vane thus offering a large surface area for beating air
- Has a strong rachis to provide firm attachment for vanes.
- Has a smooth vane to provide a stream lined body which reduces friction during flight.

2. The covert feather

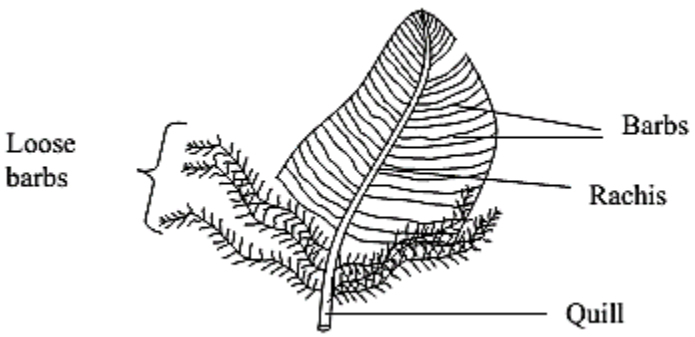
***Location*:** These are found around the neck and the upper side of the body

Characteristics:

- It is smaller than the quill feather.
- It has a large after shaft.
- It has a short vane.
- It has a soft quill.

Function: They help in temperature regulation by covering the body to prevent heat loss. They also give the body shape and colour.

Drawing of the covert feather:



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Adaptations to its functions

- Has a curved surface which gives the bird its shape.
- Has a fluffy after shaft to insulate the body.
- Have interlocking barbs and a smooth vane that make the bird water proof.

Similarities between quill and contour feathers

- Both have a quill
- Both have barbs at the base of the rachis
- Both have vanes inter locked by barbs

Differences

Quill feathers	Contour feathers
Has a stiff rachis	Has a flexible rachis
Has a long vane	Has a short vane
Has a long quill	Has a short quill
Barbs closely interlocked	Less interlocked barbs

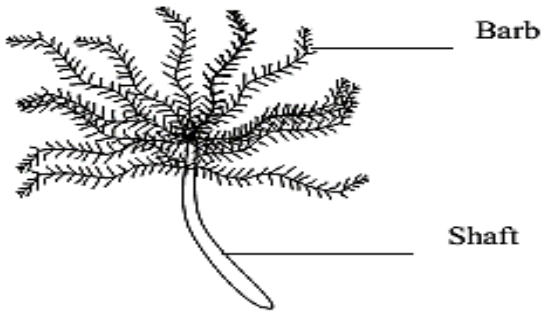
3. Down feathers

***Location*:** These are found all over the body especially on the abdomen and between covert and flight feathers.

Characteristics:

- It is smaller than the quill and covert feathers.
- It is soft and fluffy.
- It has a short and small quill.
- It has no vane but instead it has free barbs.

Structure of the down feather



Function: It insulates the body against heat loss.

4. Filo plumes

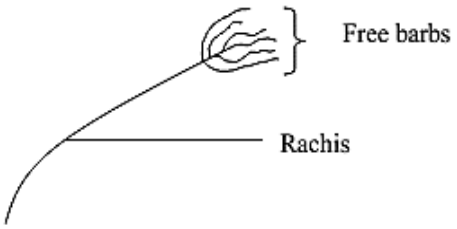
***Location*:** These are found sparsely distributed all over the body amongst other feathers.

Characteristics:

- They are long and hair-like feathers.
- They have a long rachis.
- They have a thread-like shape.
- They have few free barbs at one end

Function. These feathers are for sensation.

Structure of the filoplume



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